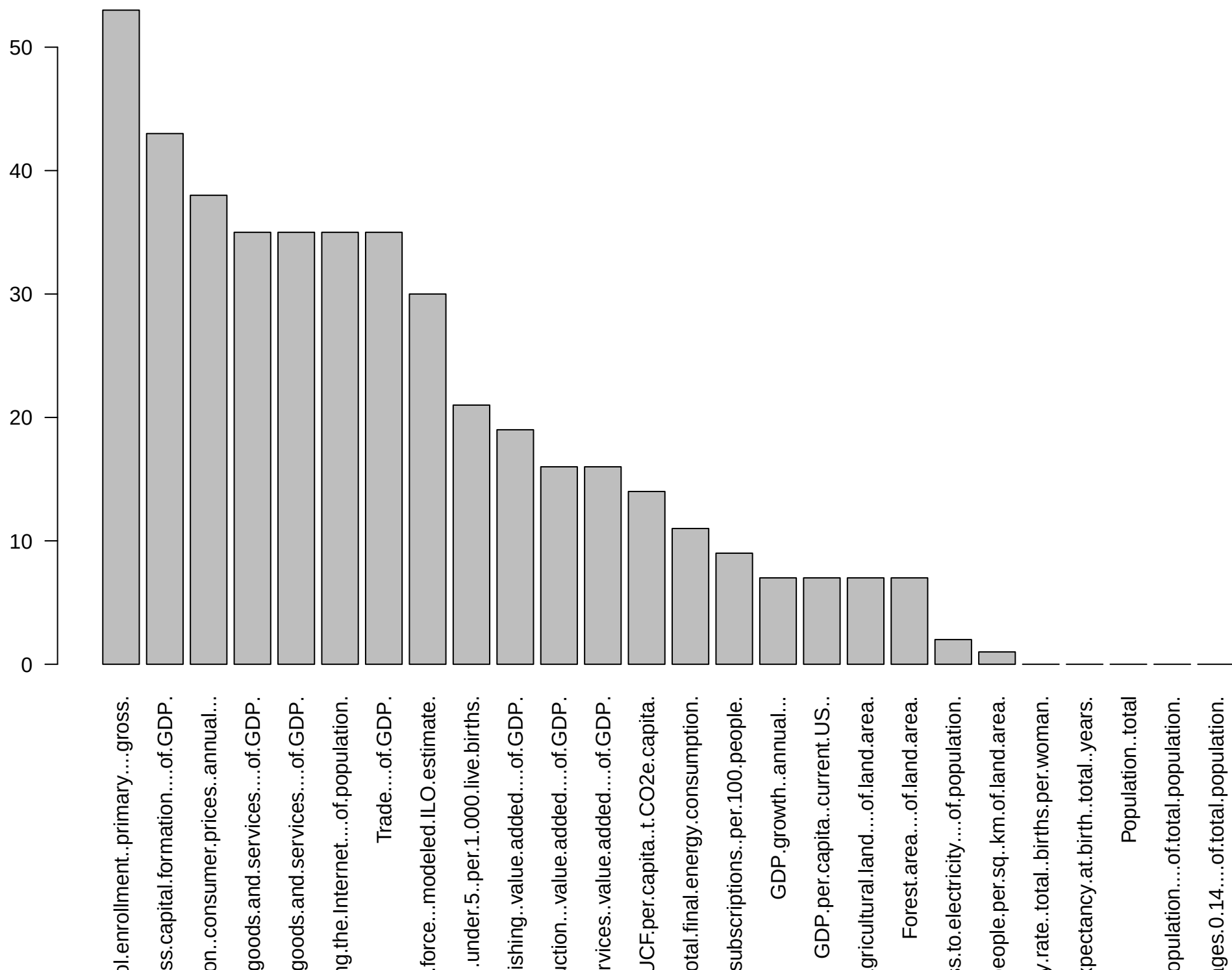
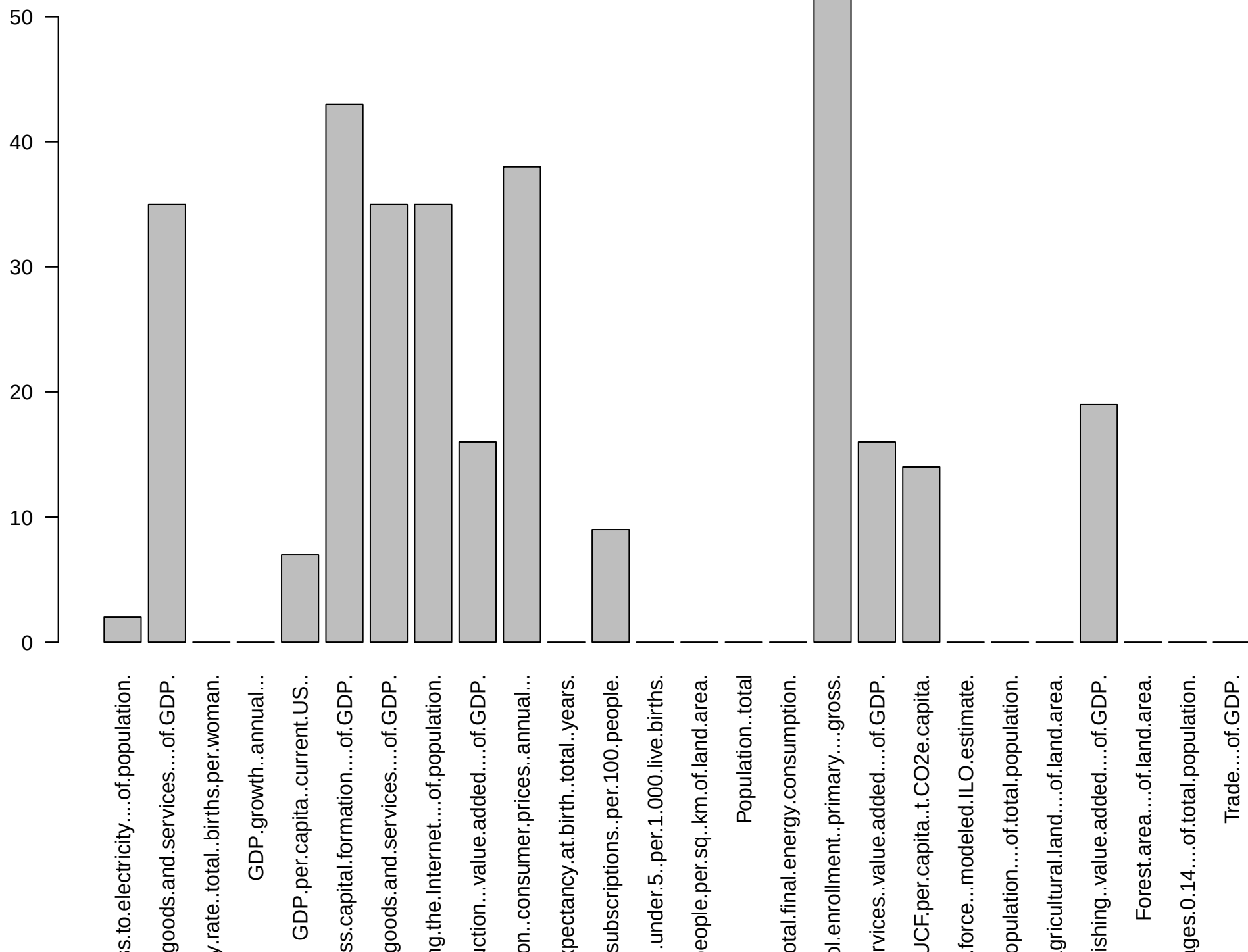


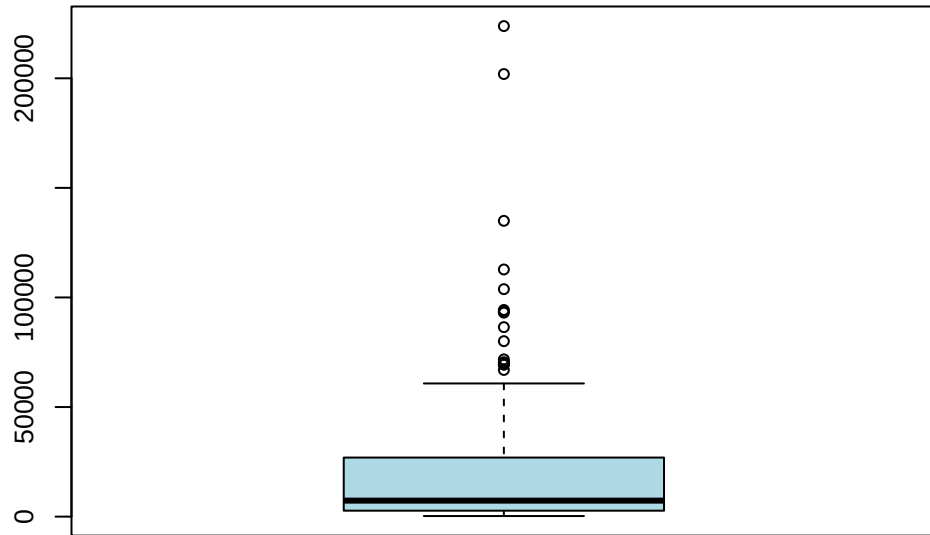
Zeros por variável (tratados como NA)



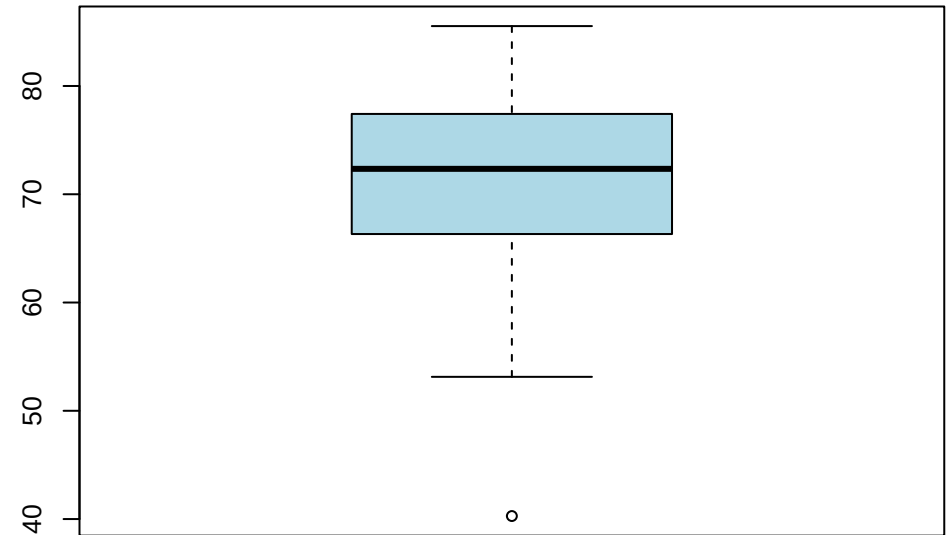
Número de missings por variável



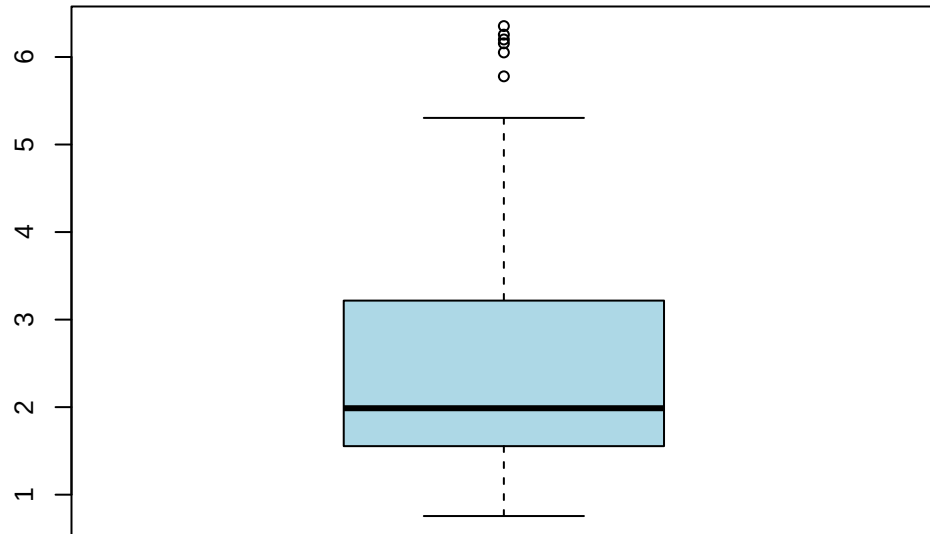
GDP per capita (USD)



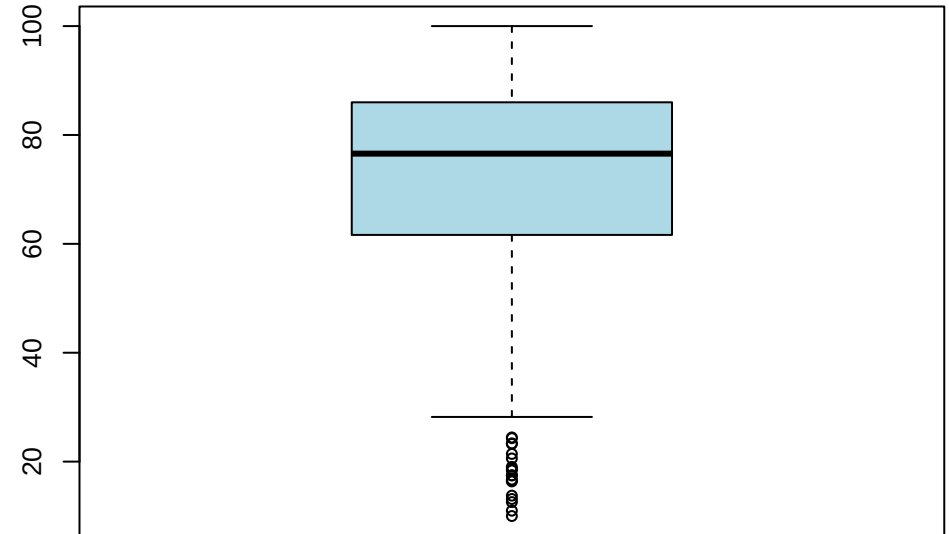
Life expectancy (years)



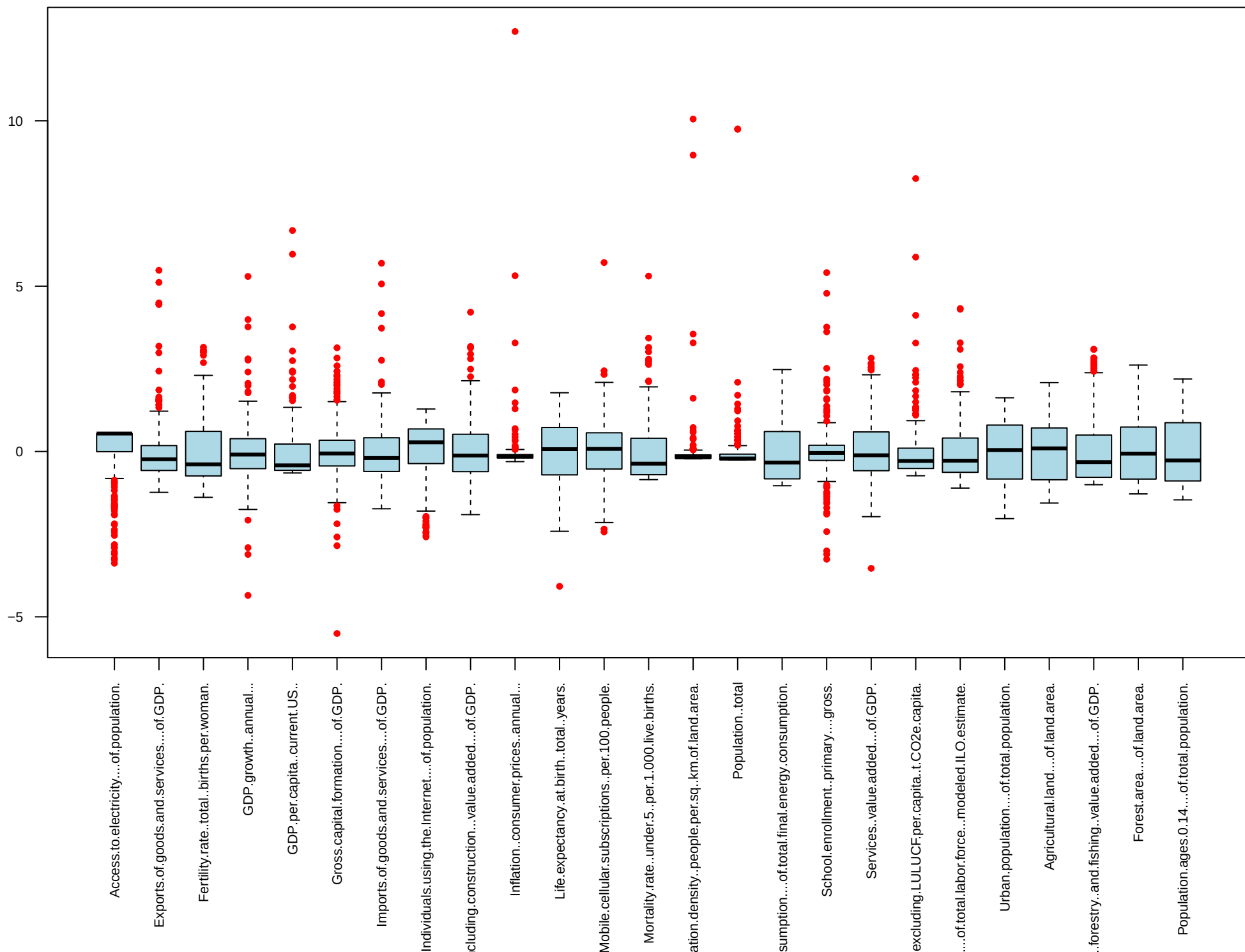
Fertility rate

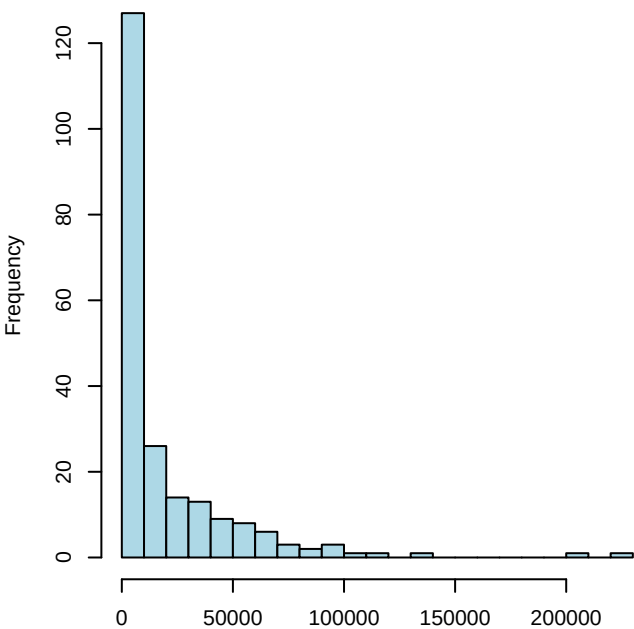
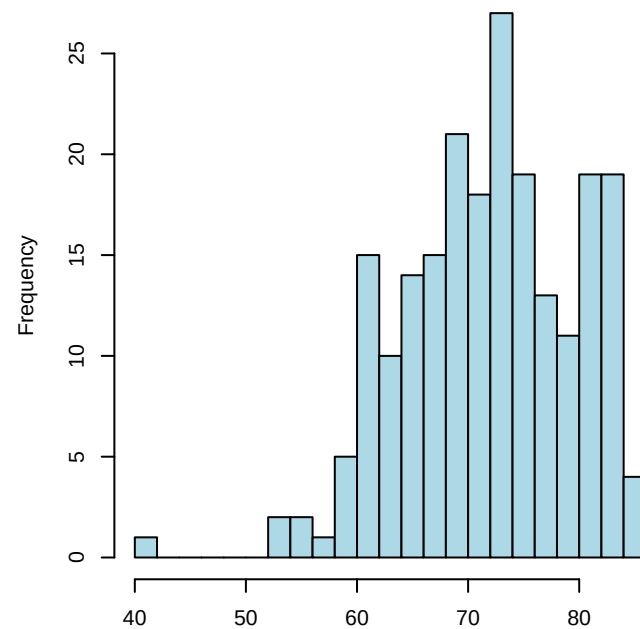
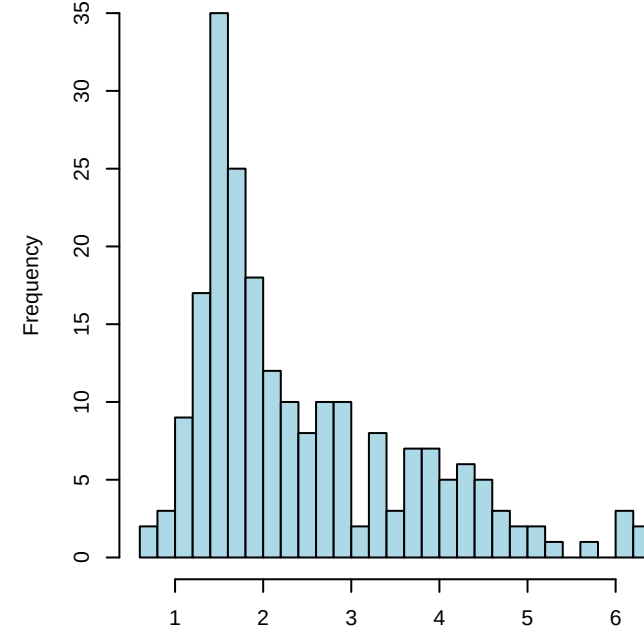
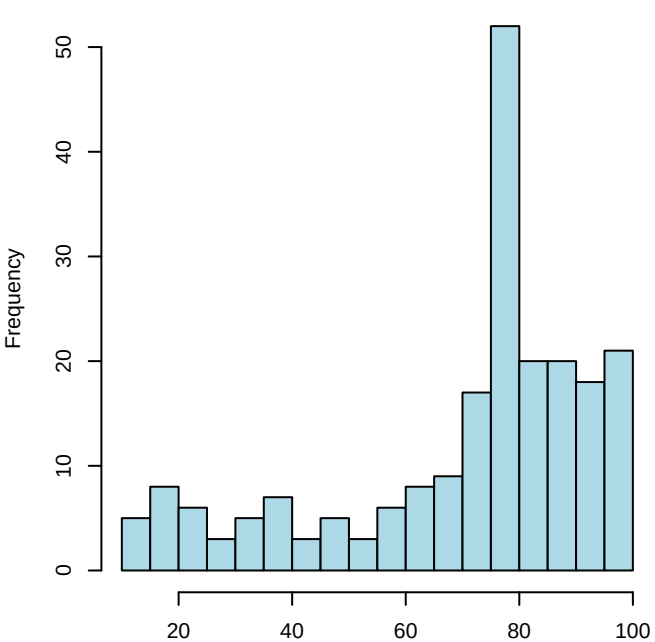
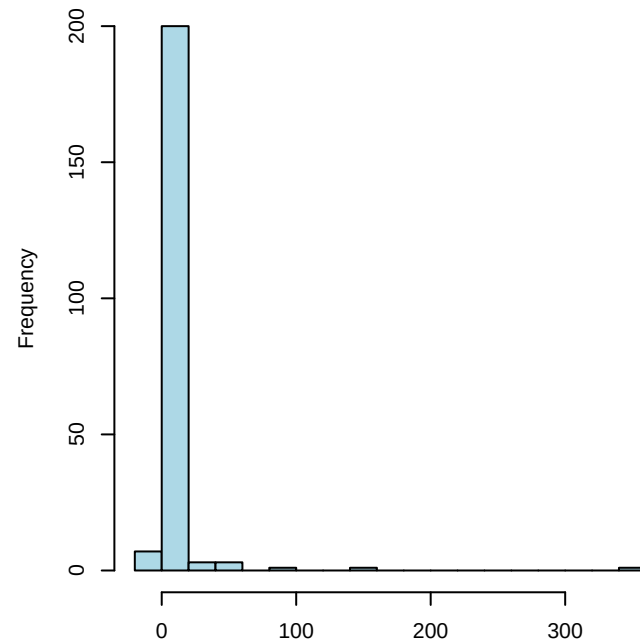
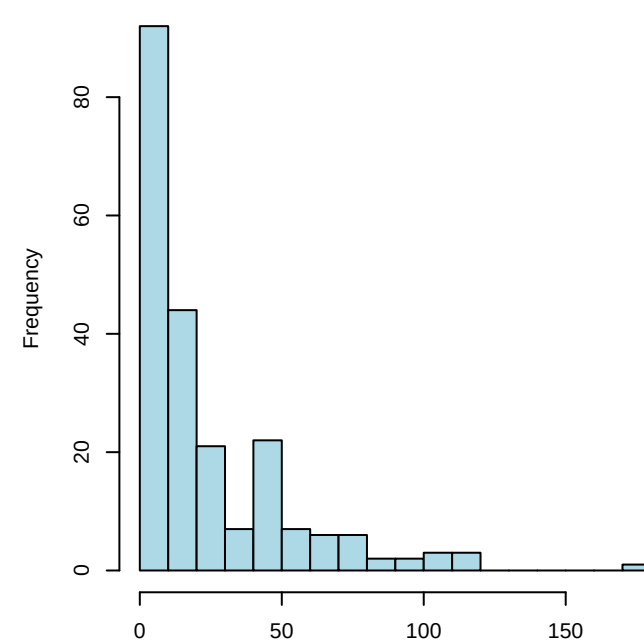


Internet users (%)

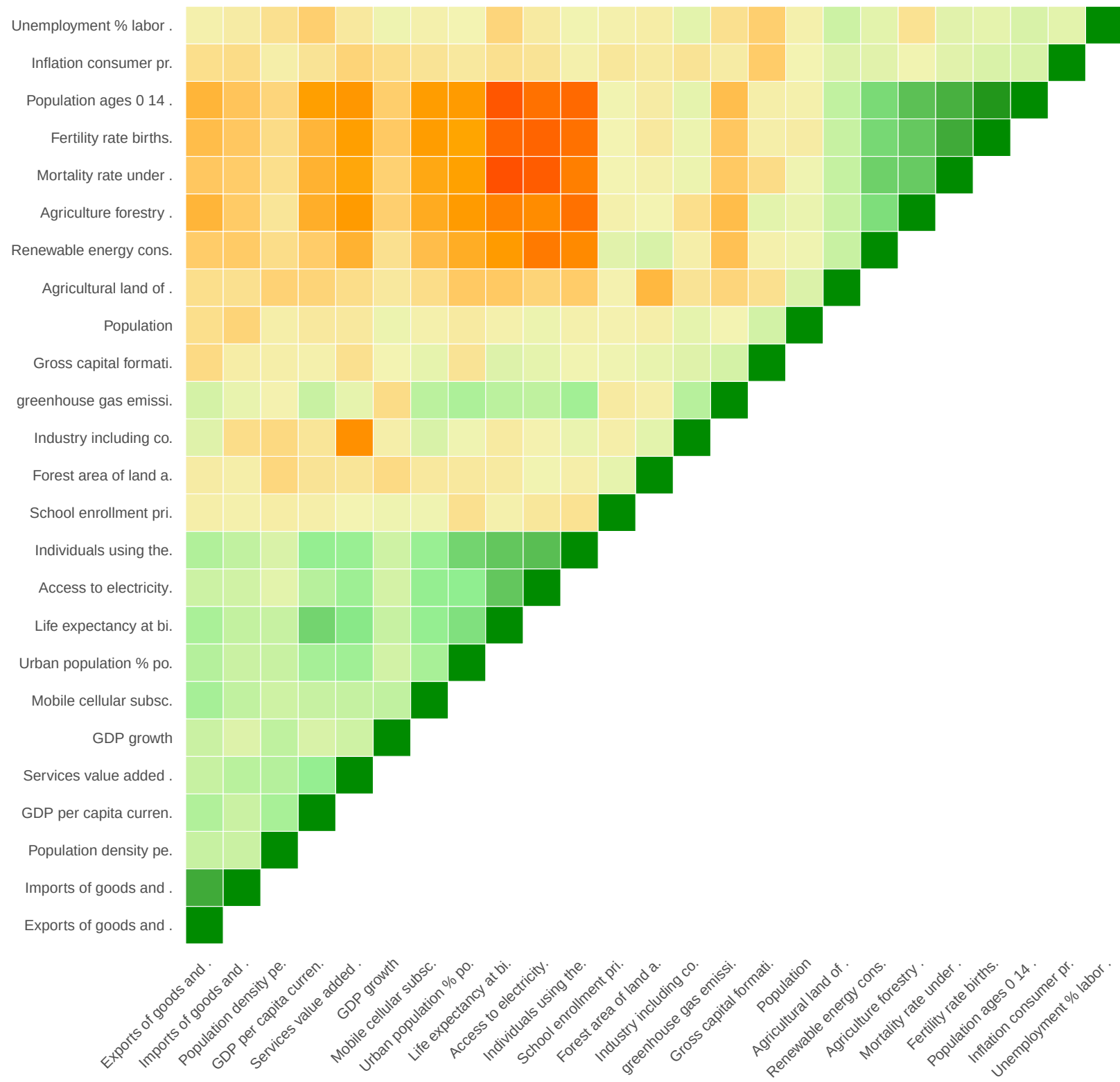


Standardised boxplots (z-score)

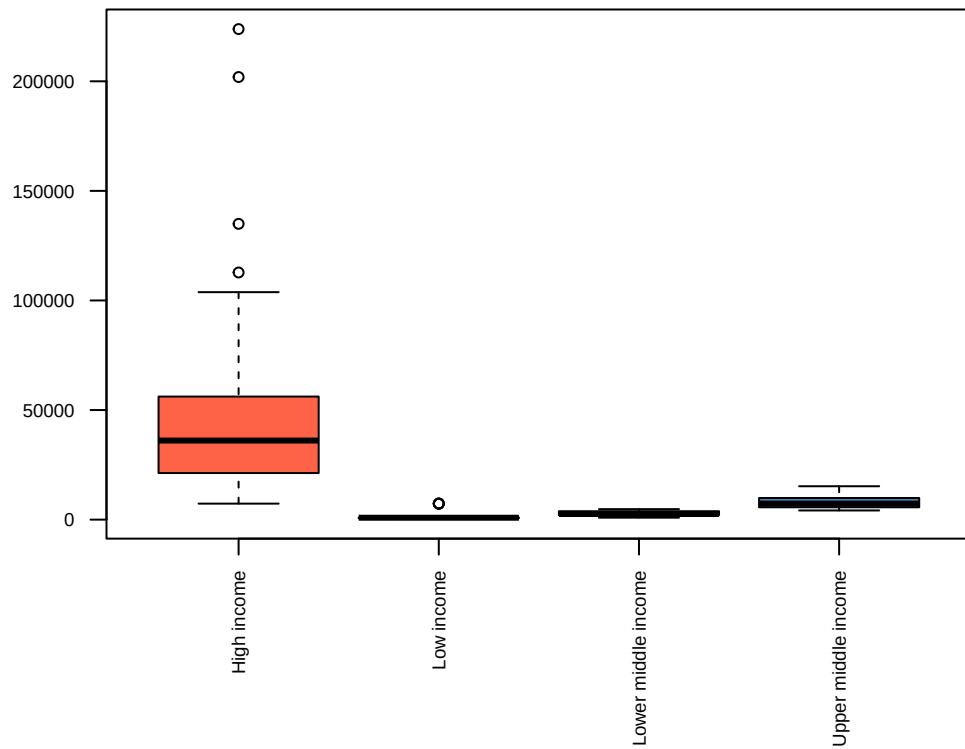


GDP per capita**Life Expectancy****Fertility Rate****Internet Users (%)****Inflation (%)****Child Mortality**

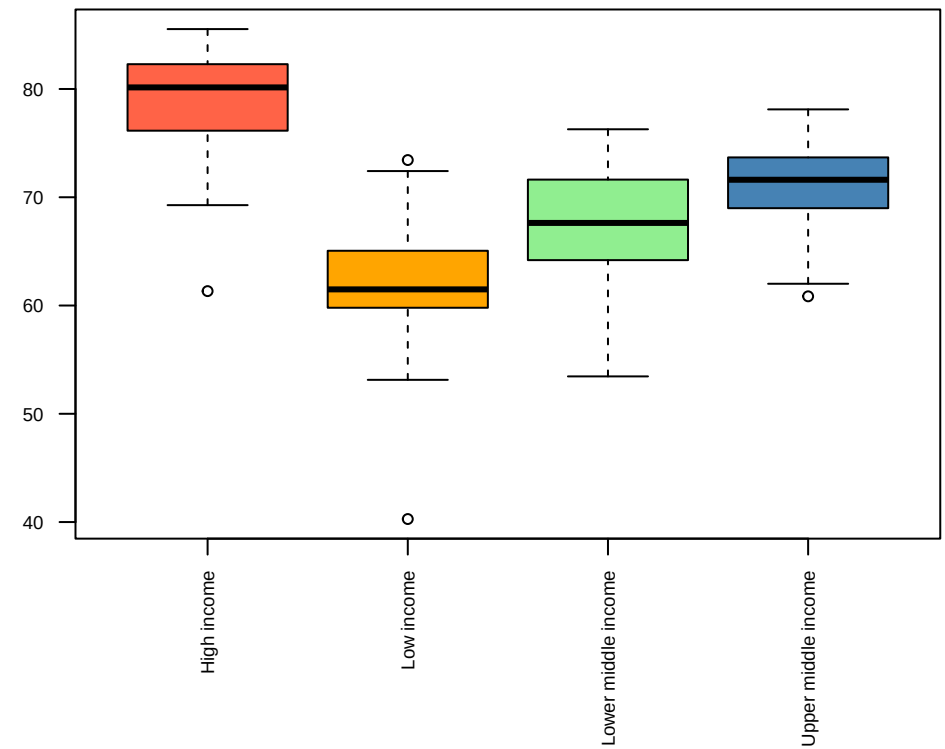
Correlation heatmap



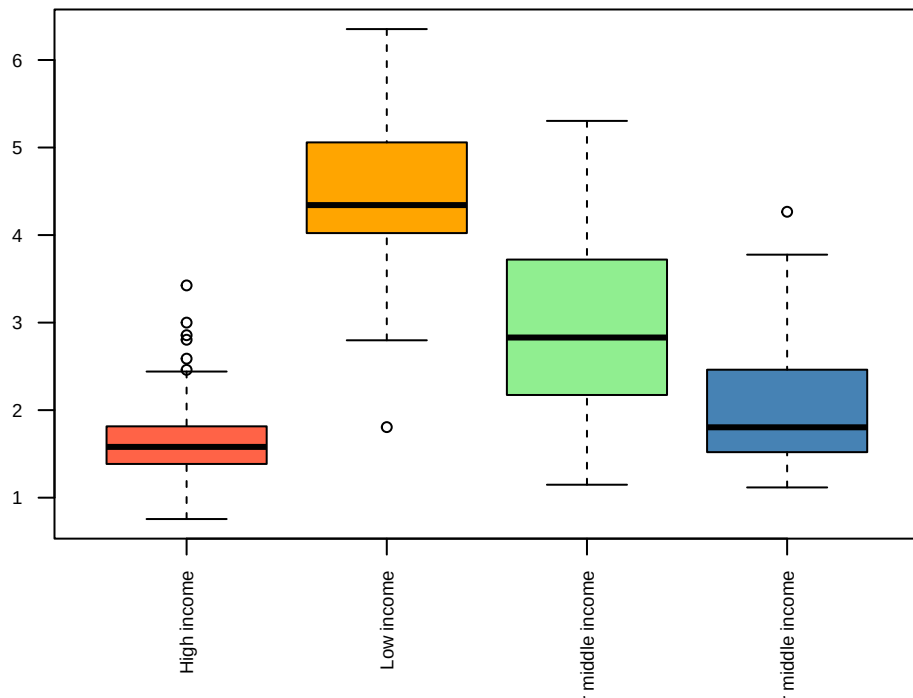
GDP per capita (USD)



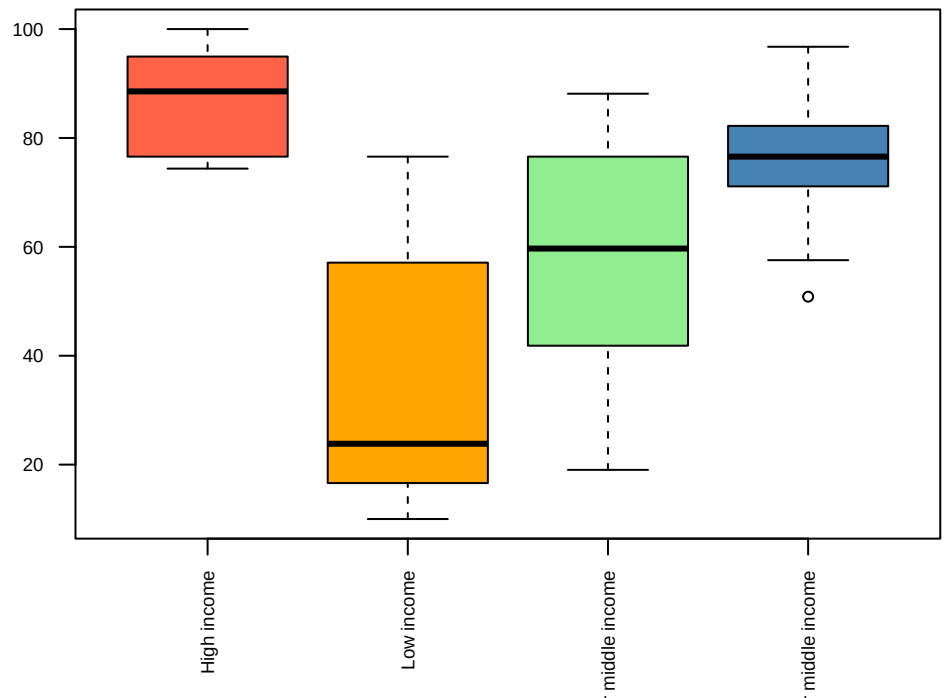
Life expectancy (years)



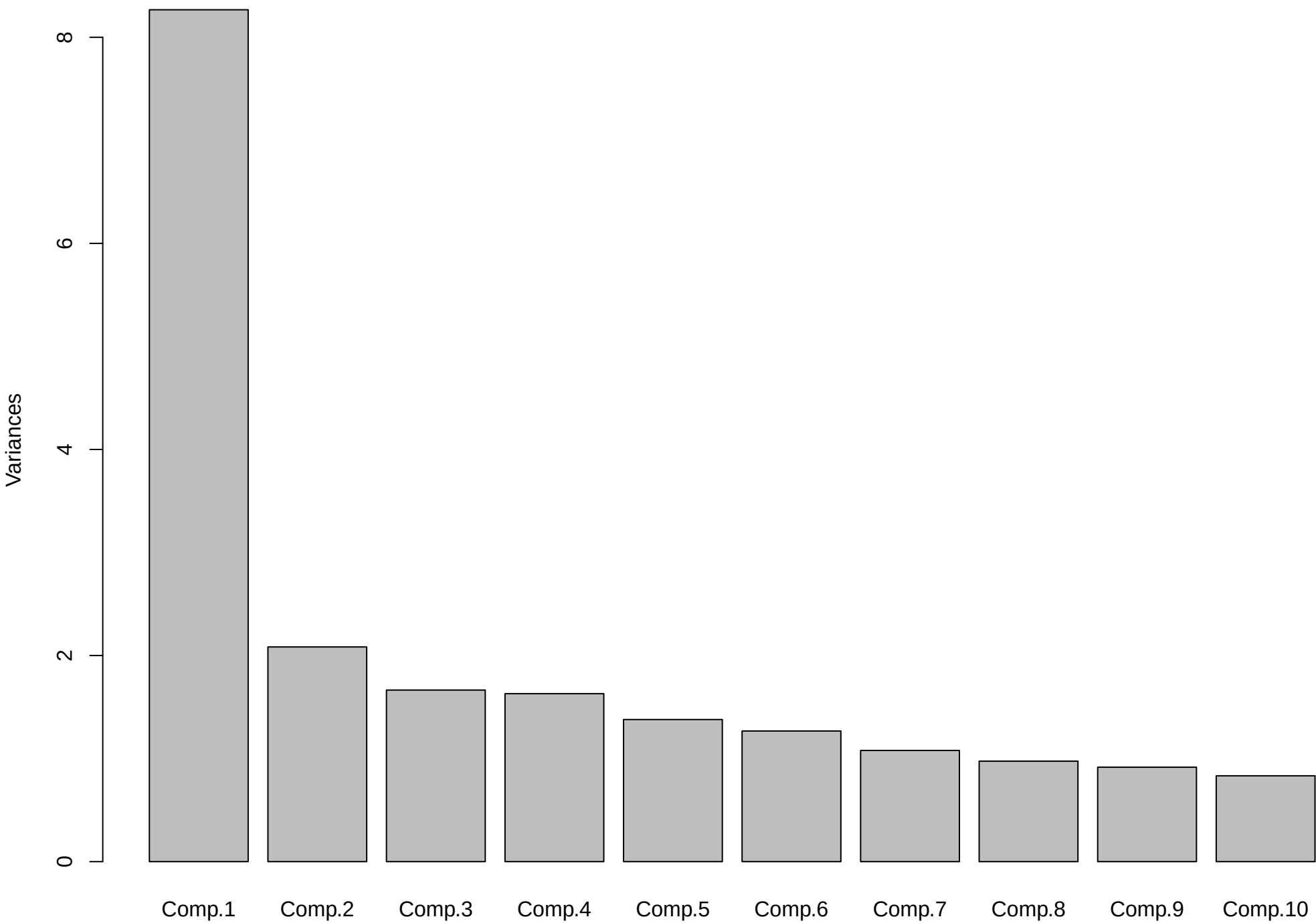
Fertility rate



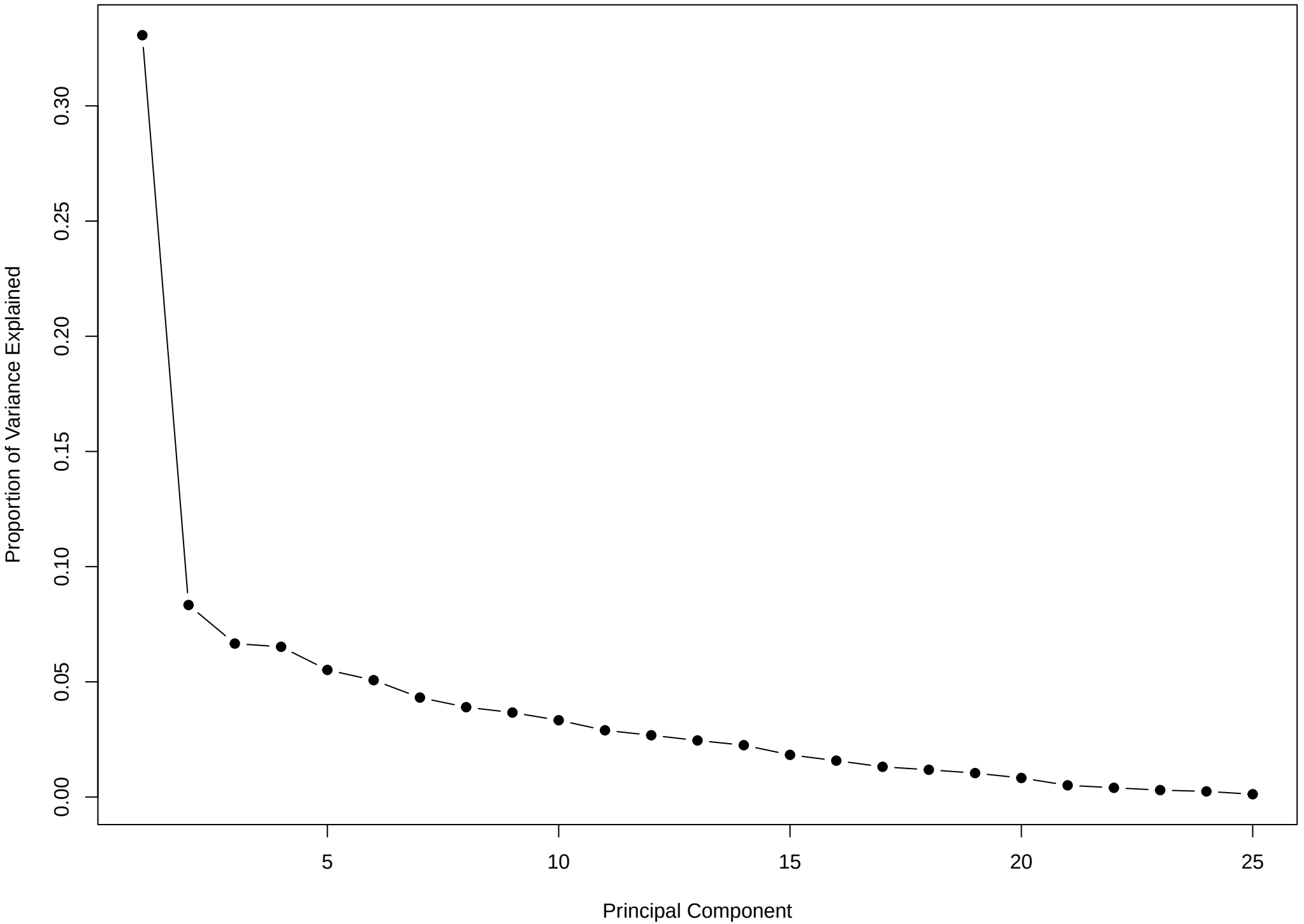
Internet users (%)



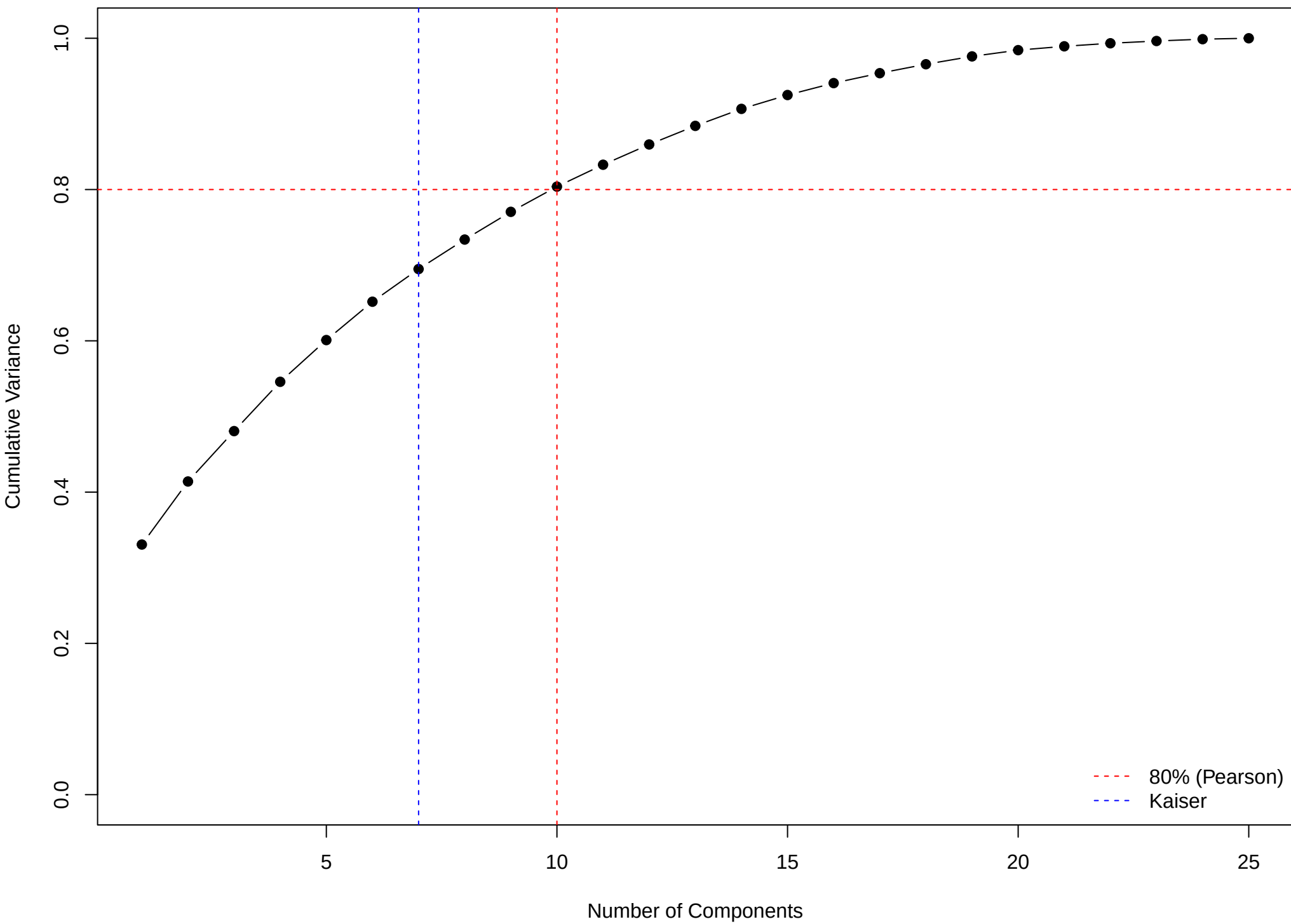
Variance per component (Scree Plot)



Scree Plot



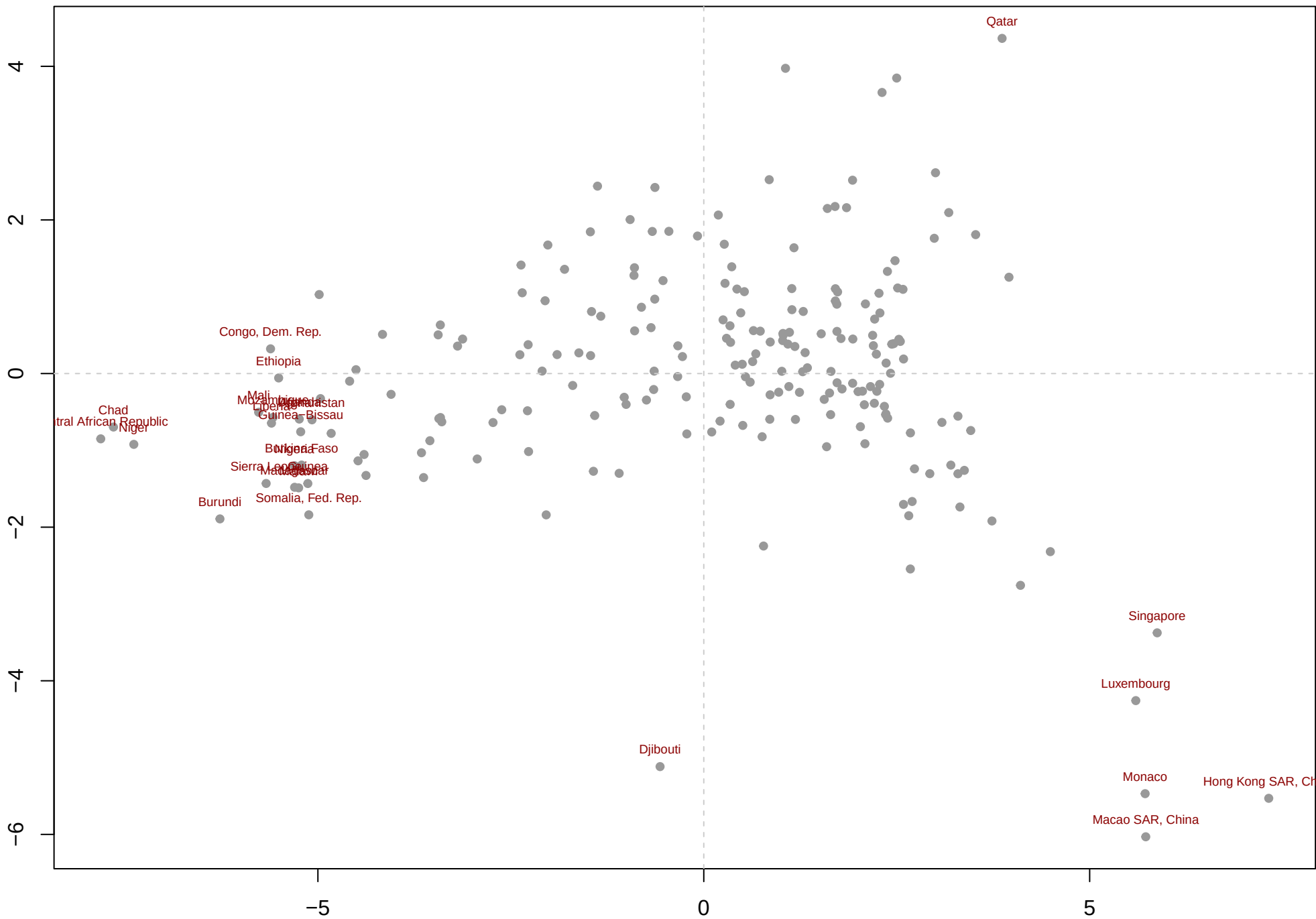
Cumulative Variance



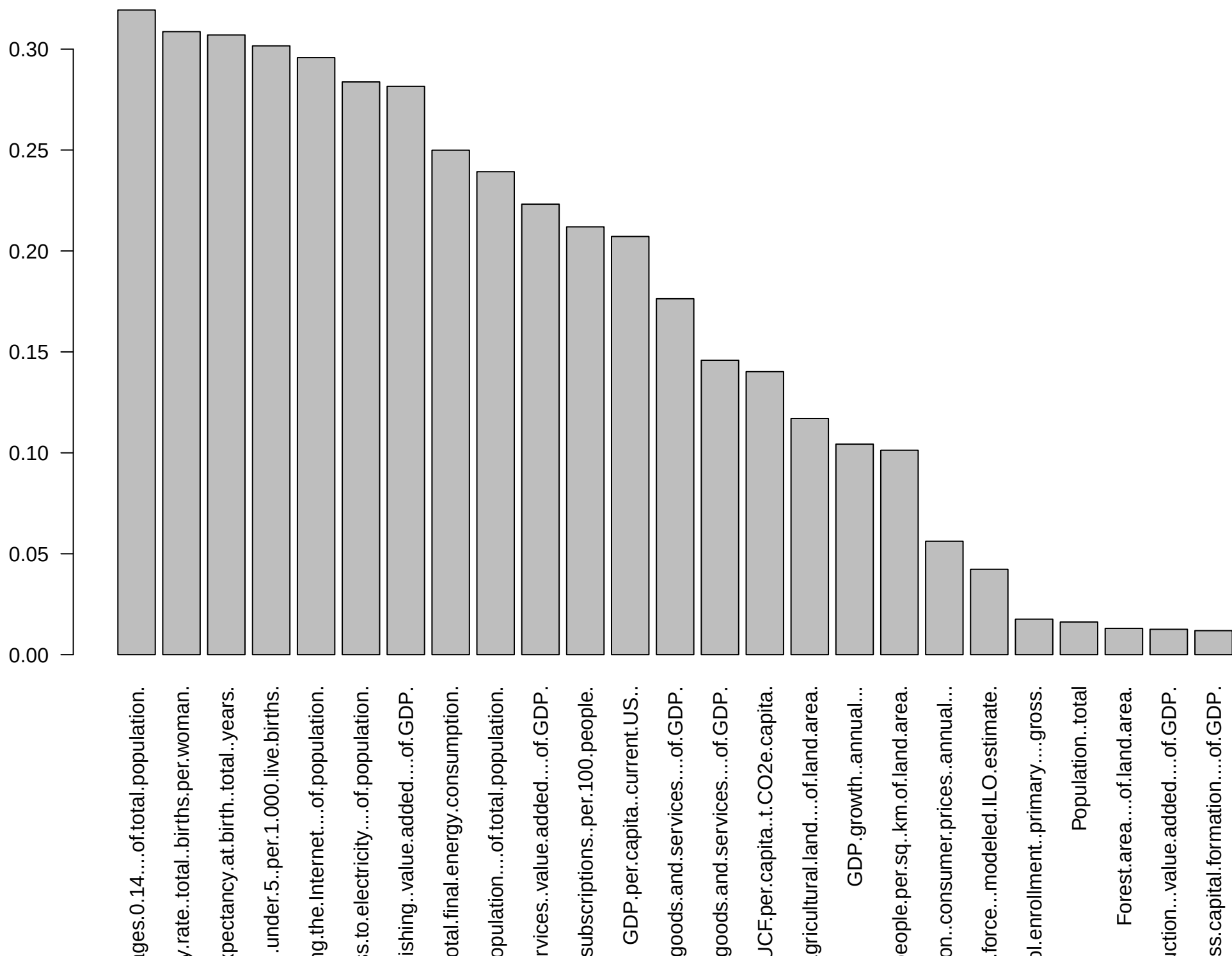
Scores PC1 vs PC2

PC2 – Productive Structure (8.33%)

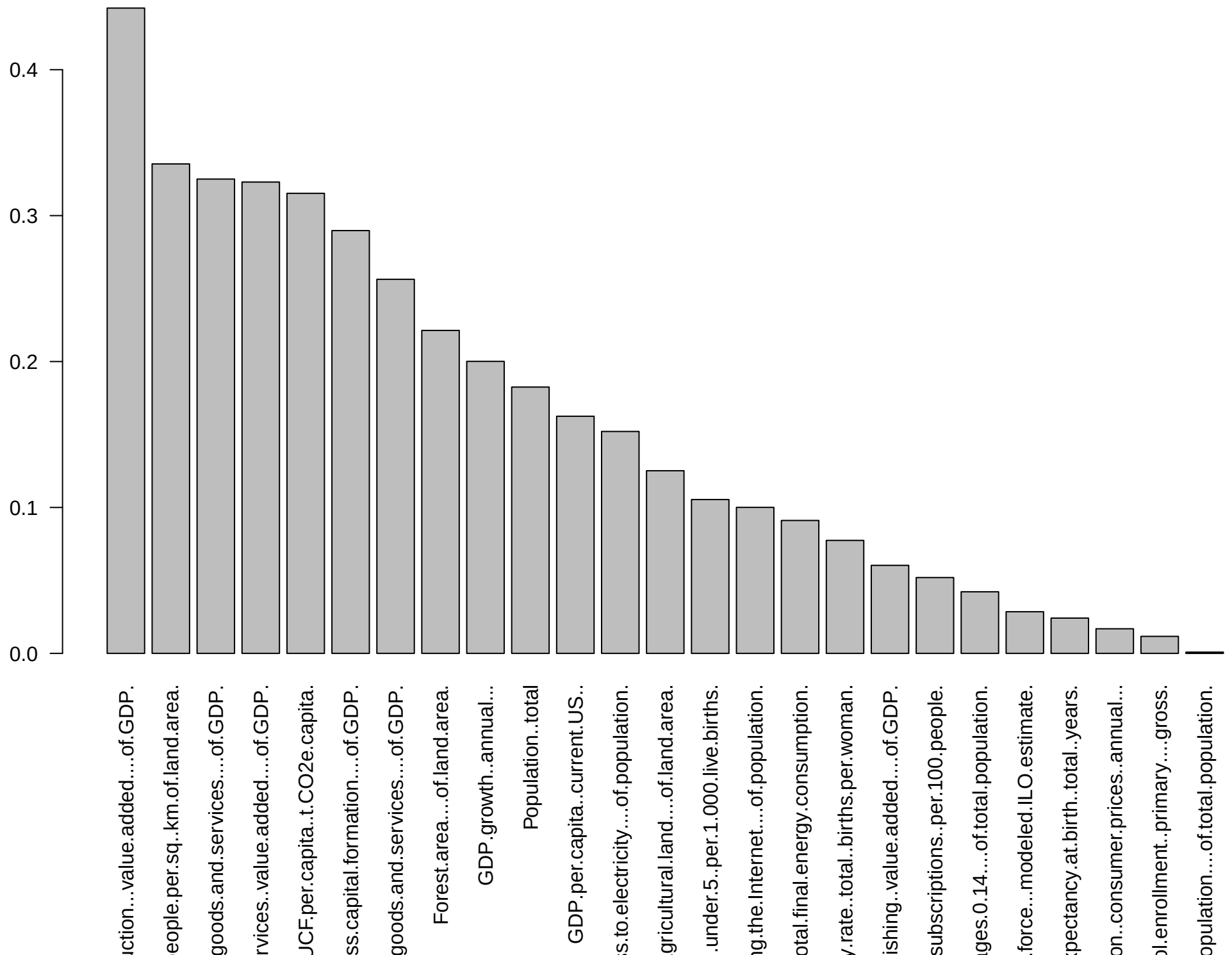
PC1 – Human Development (33.07%)



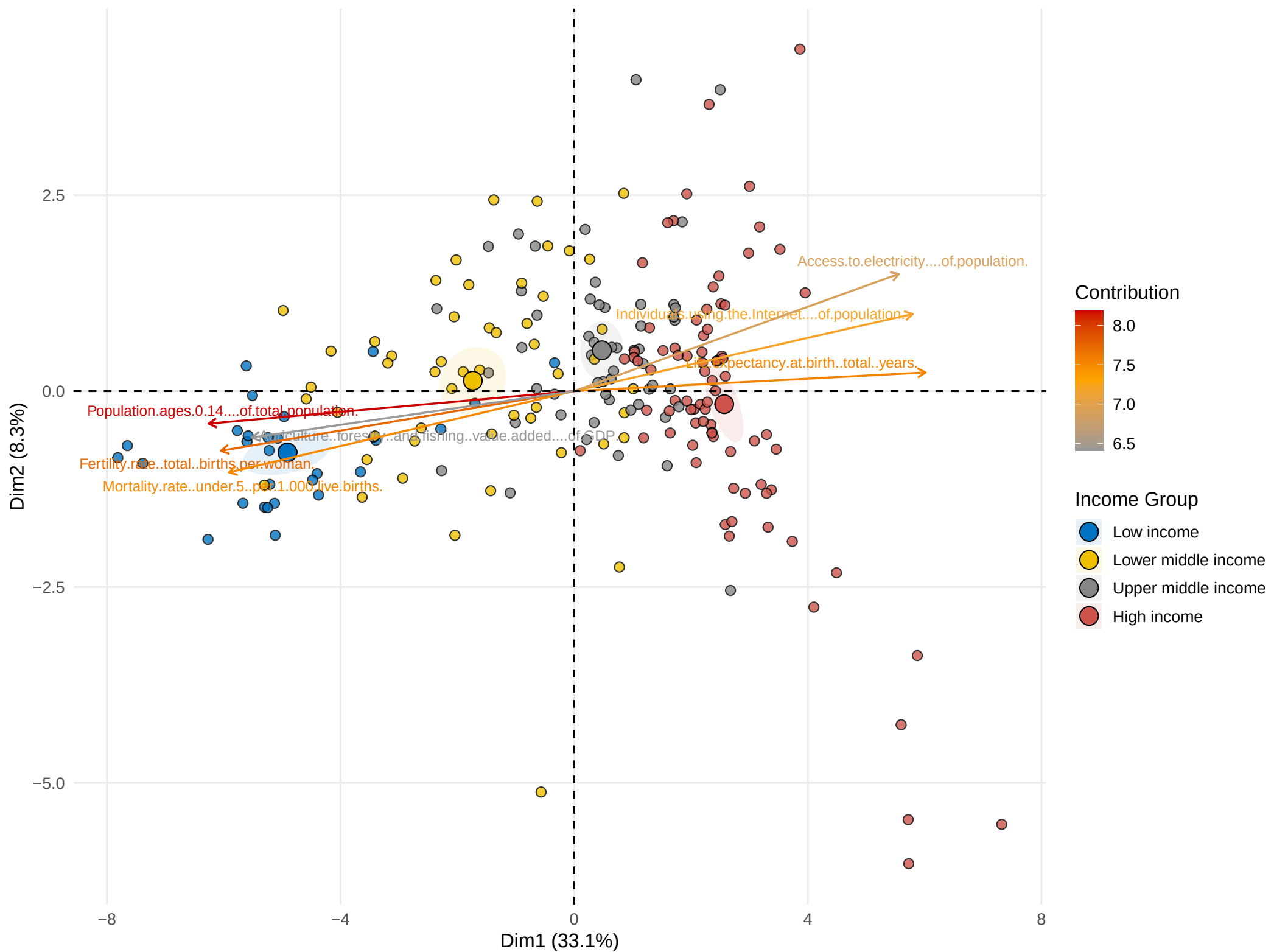
Variable importance on PC1 (Human Development)



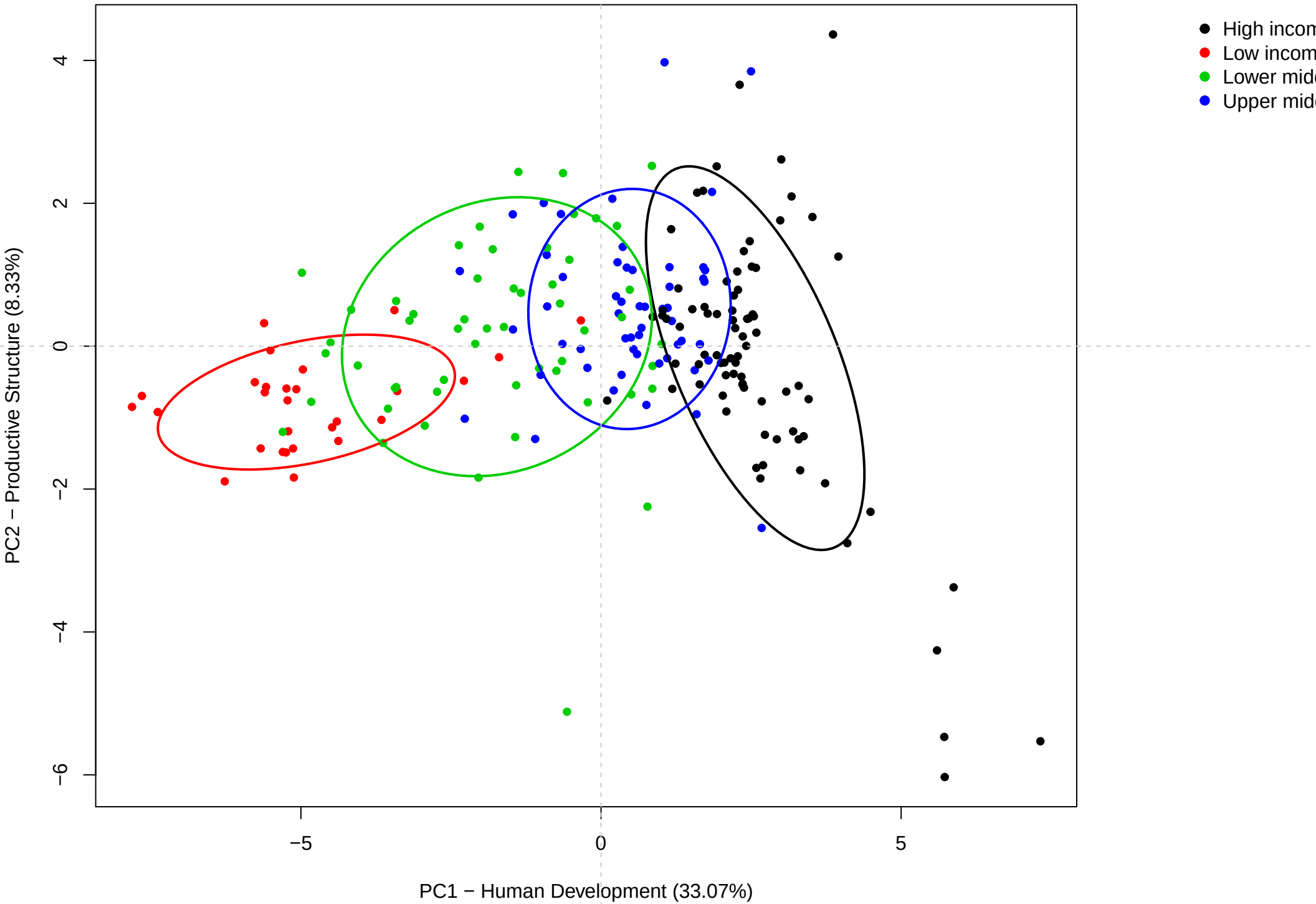
Variable importance on PC2 (Productive Structure)



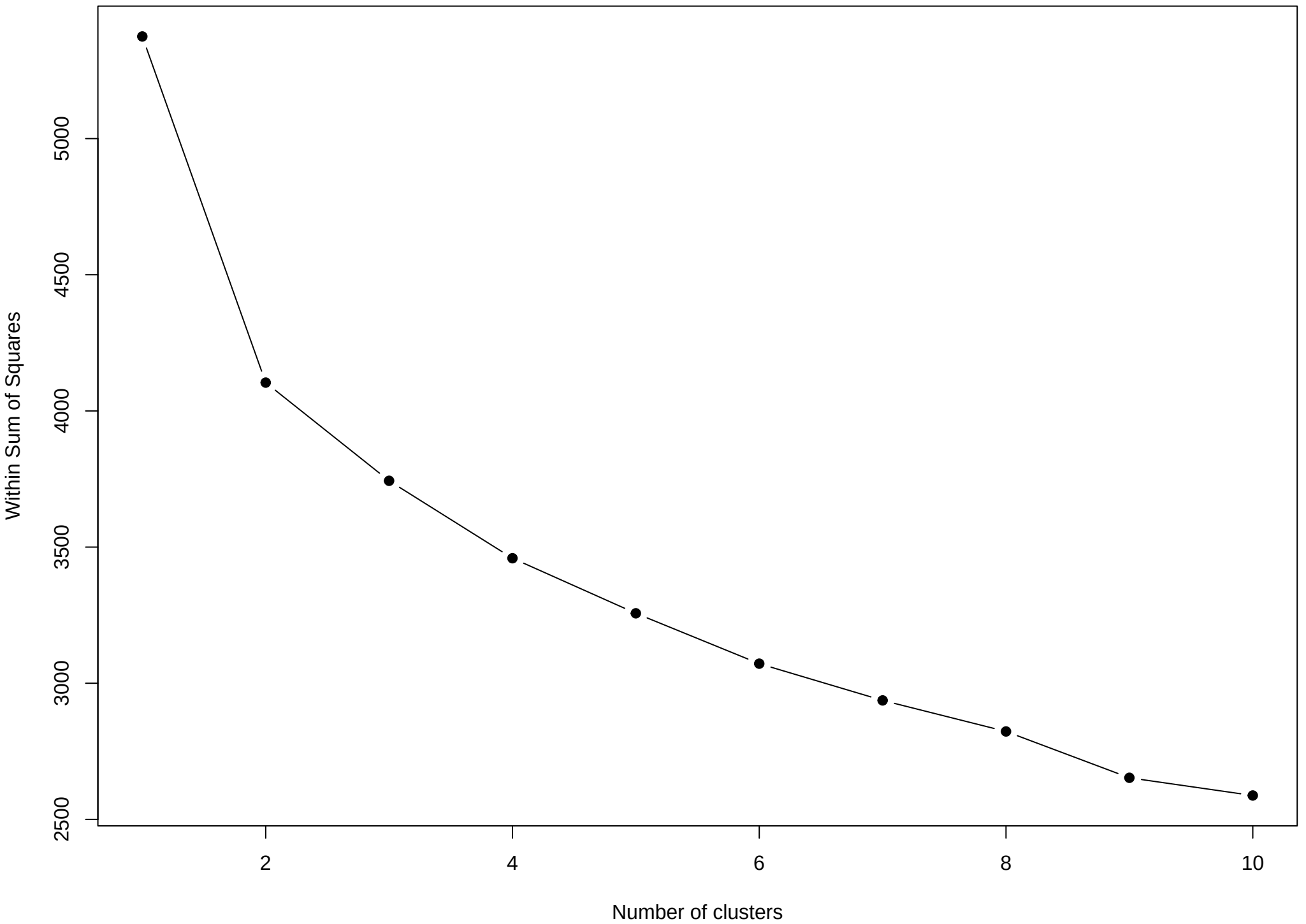
Biplot – PCA (PC1 vs PC2)



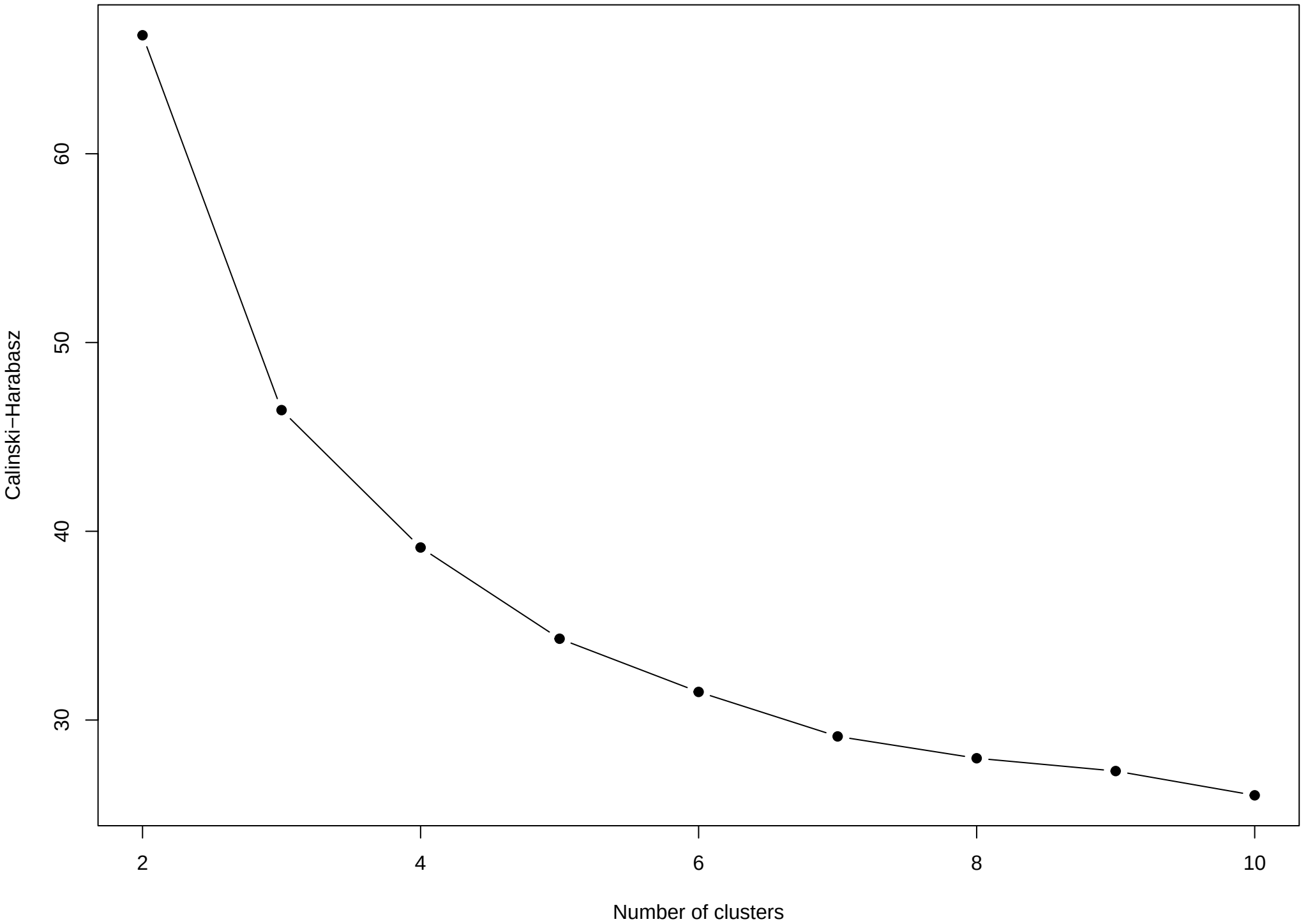
PCA by Income Group



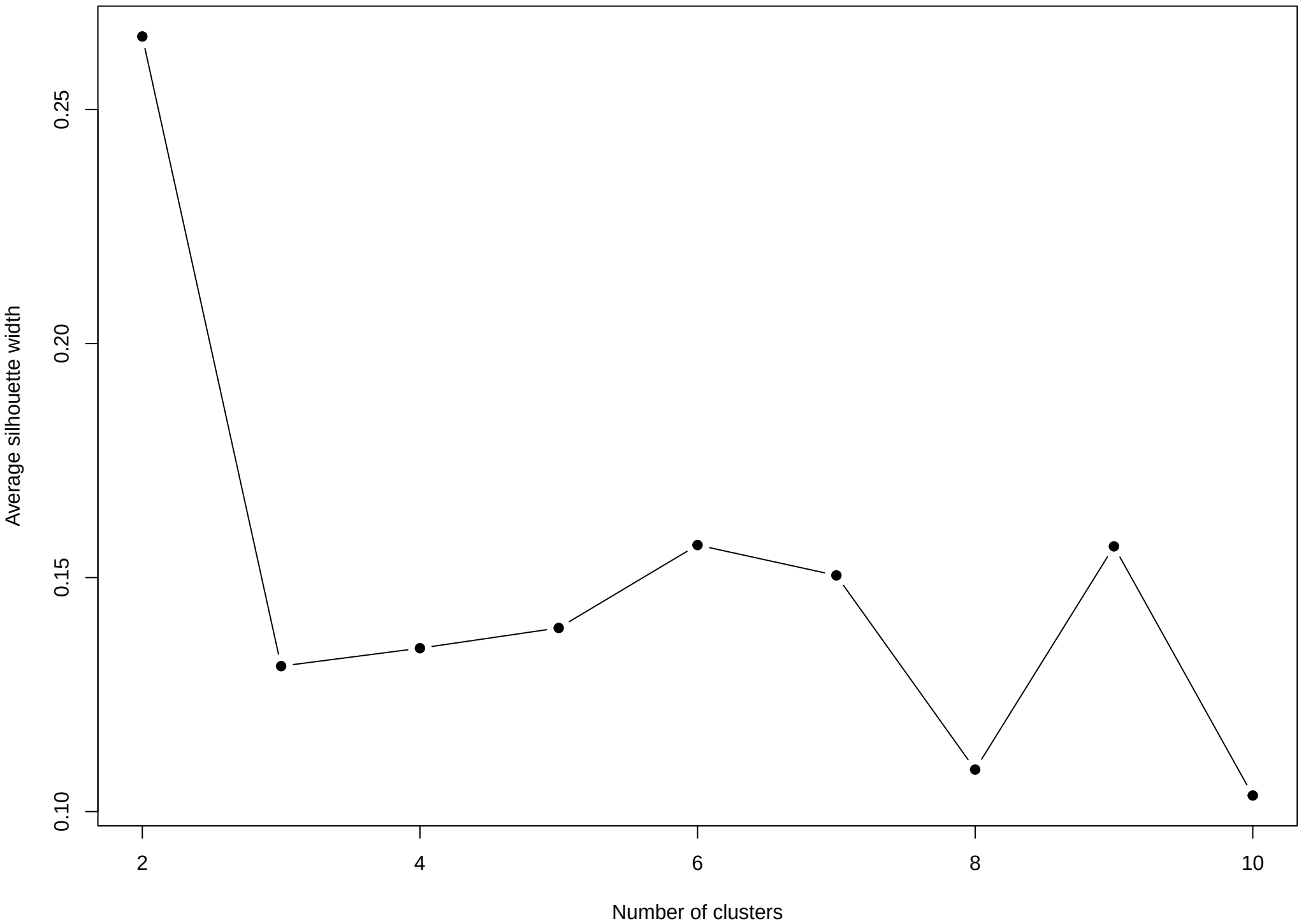
Elbow Method



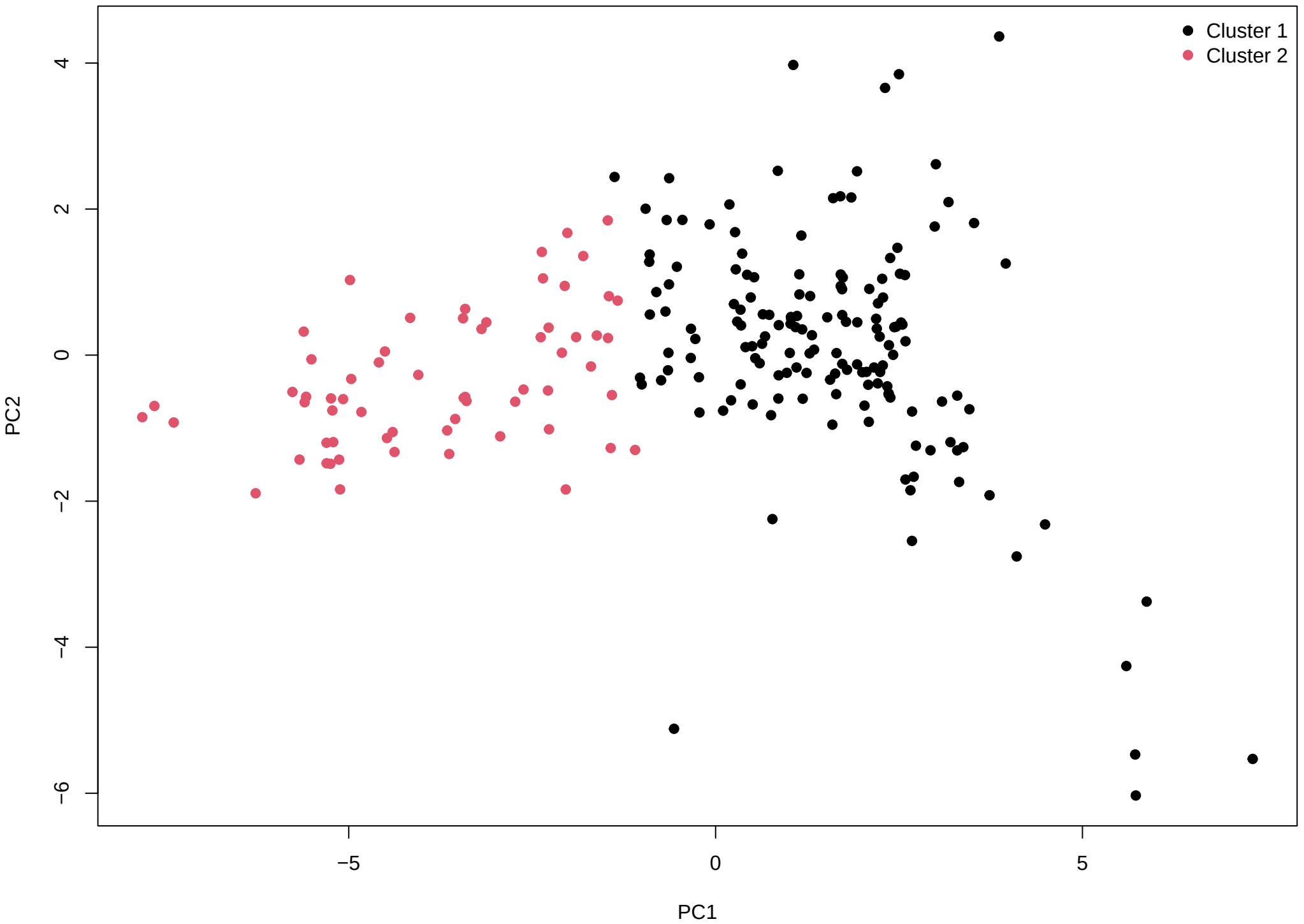
Calinski-Harabasz Criterion



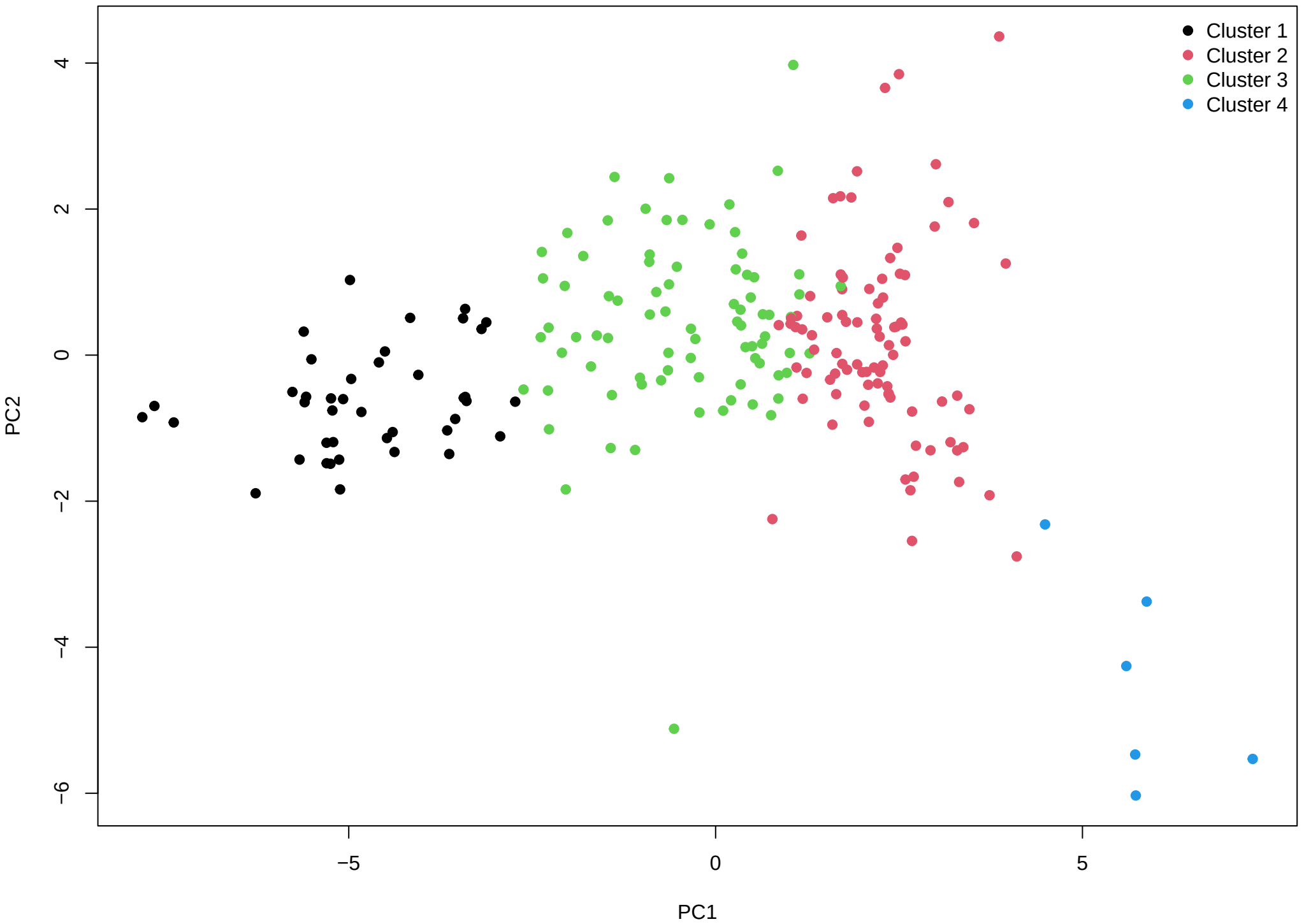
Silhouette Method



K-Means with K = 2

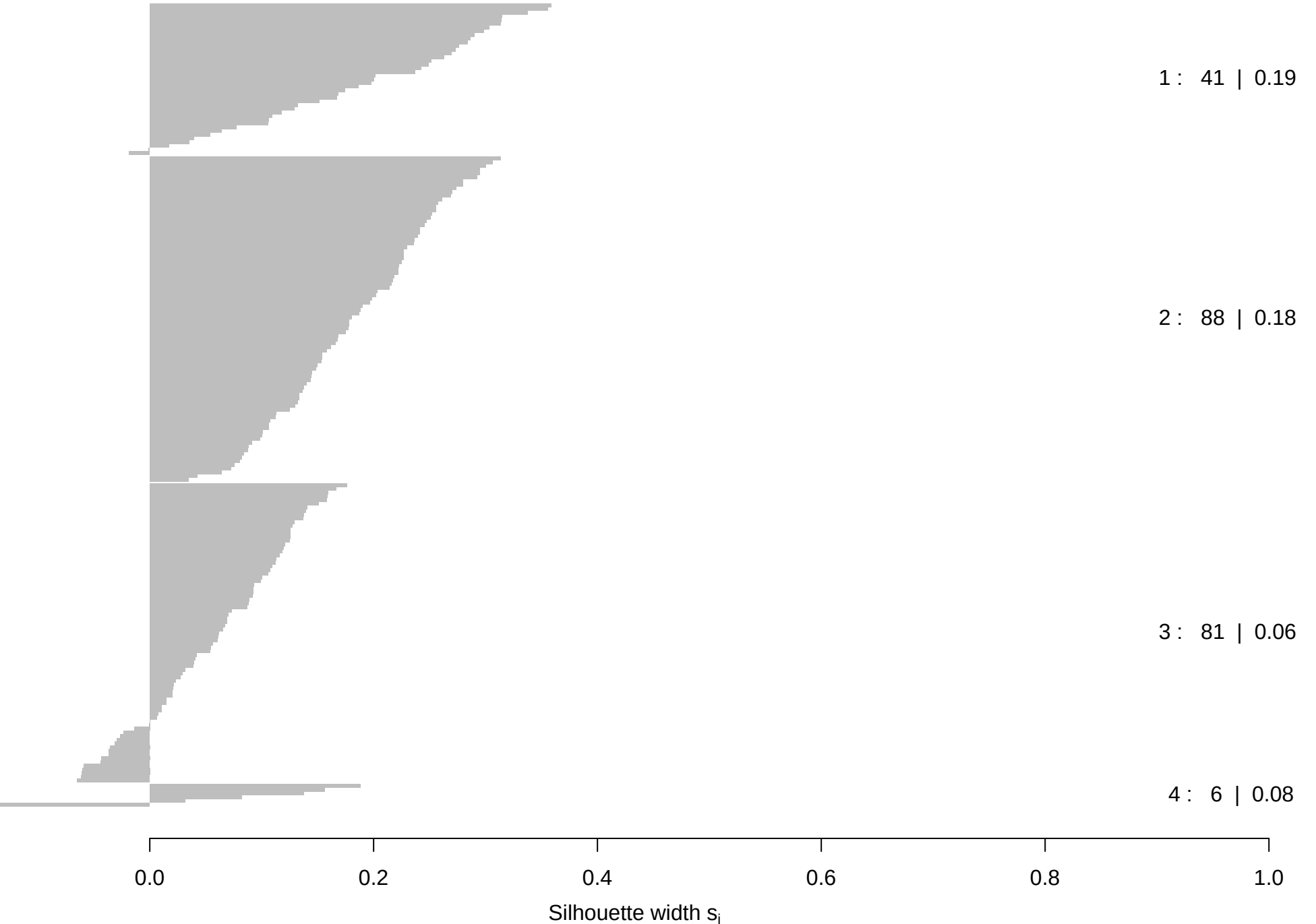


K-Means with K = 4



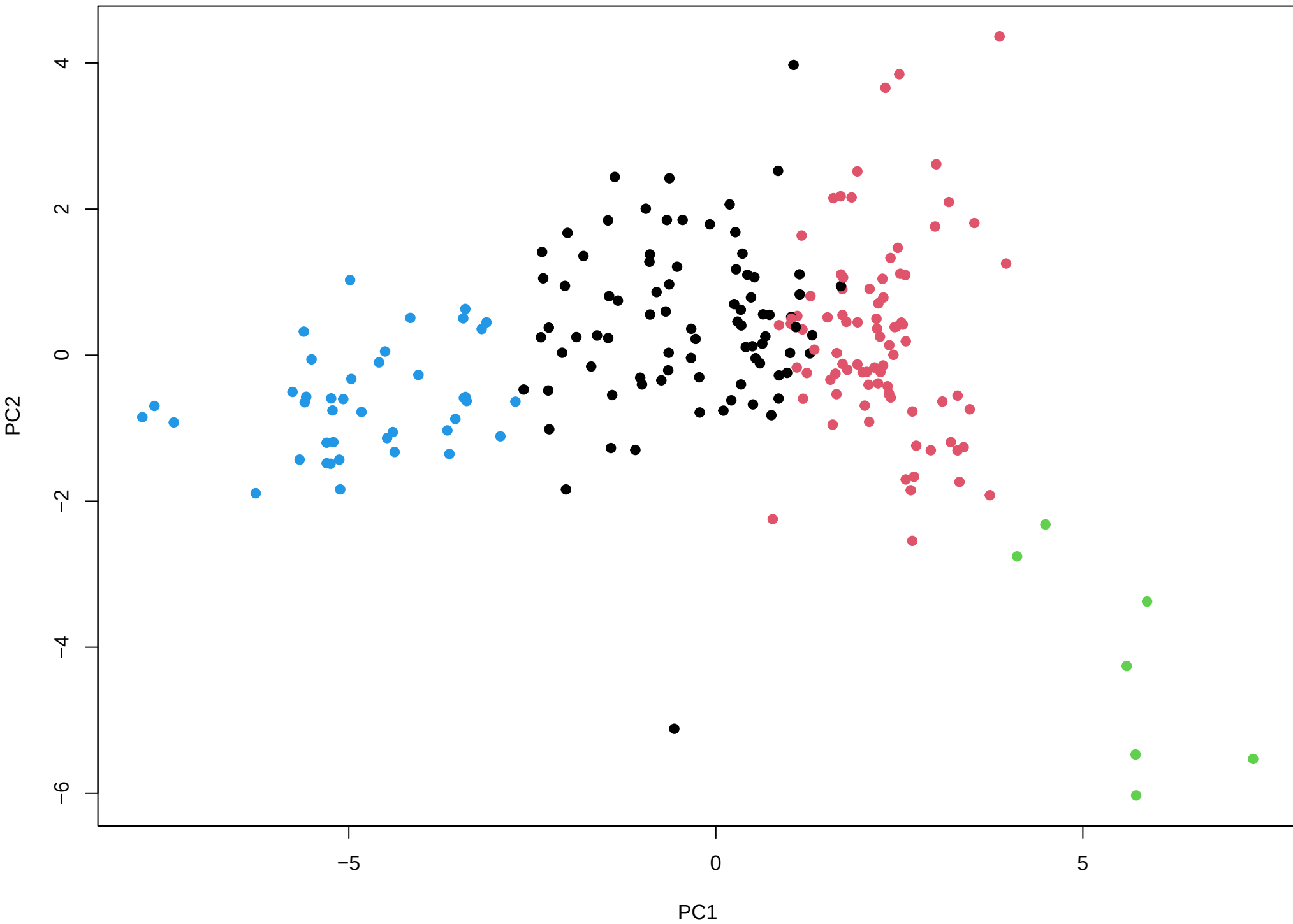
Silhouette K = 4

n = 216

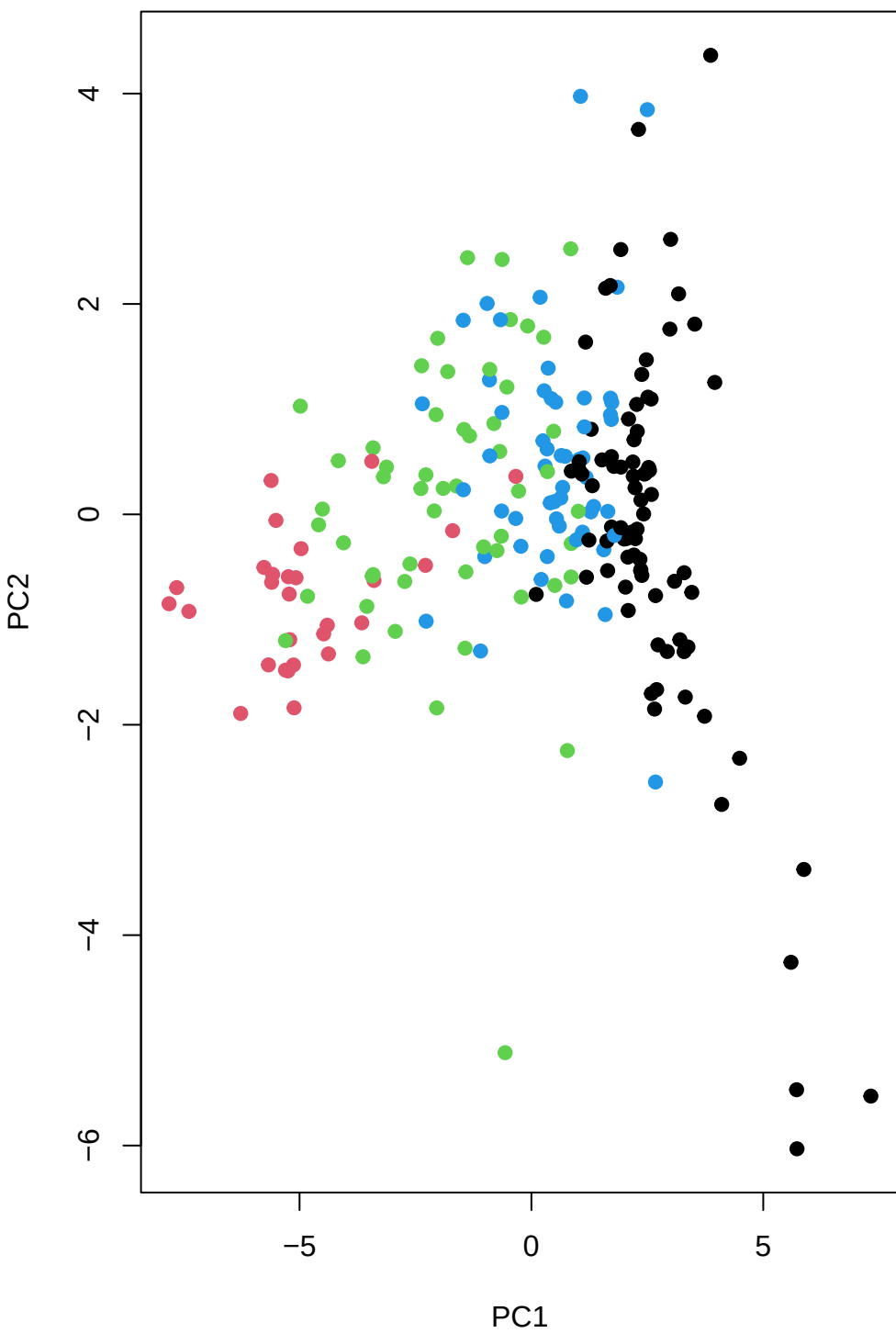


Average silhouette width : 0.13

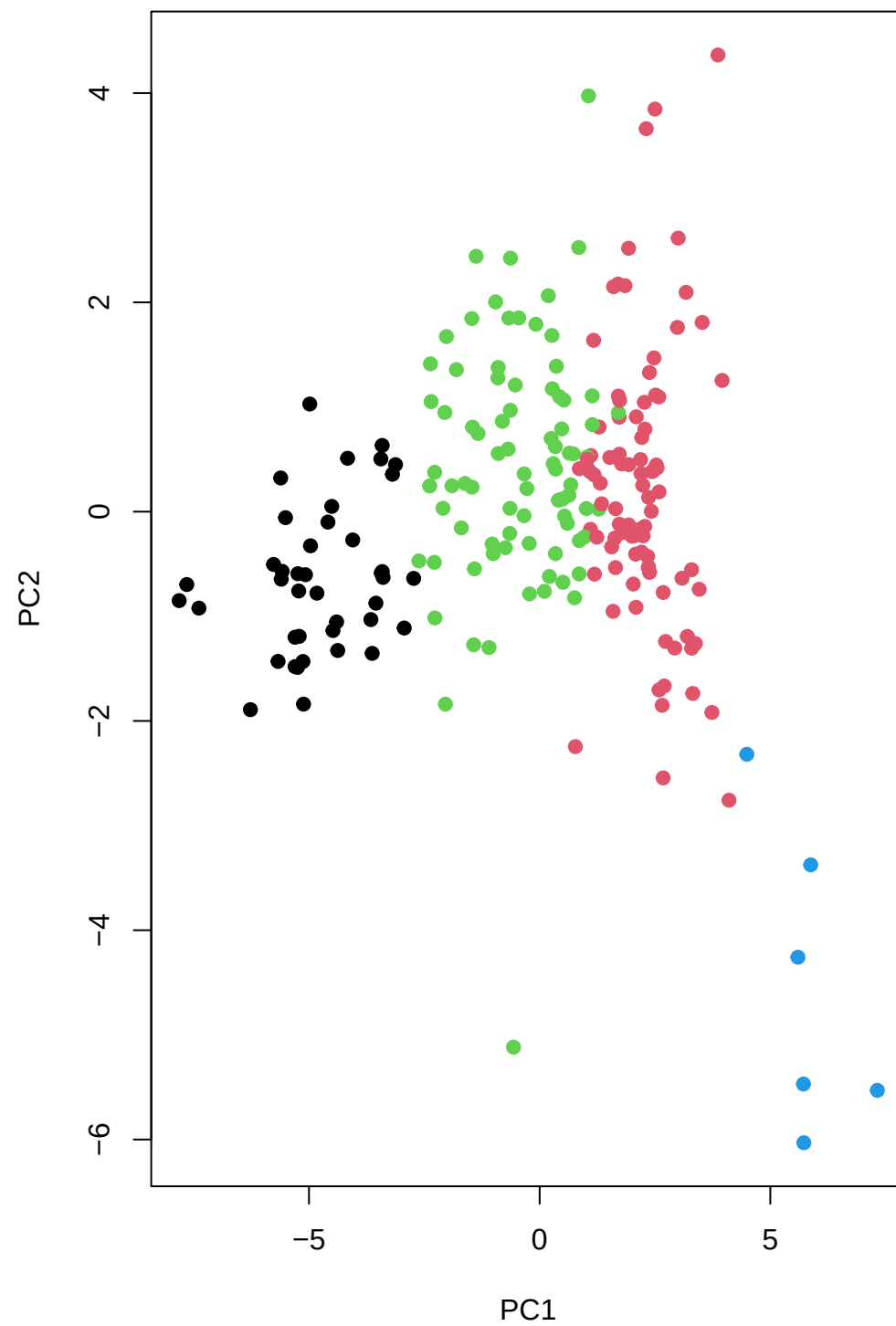
K-Means on PCs (K=4)

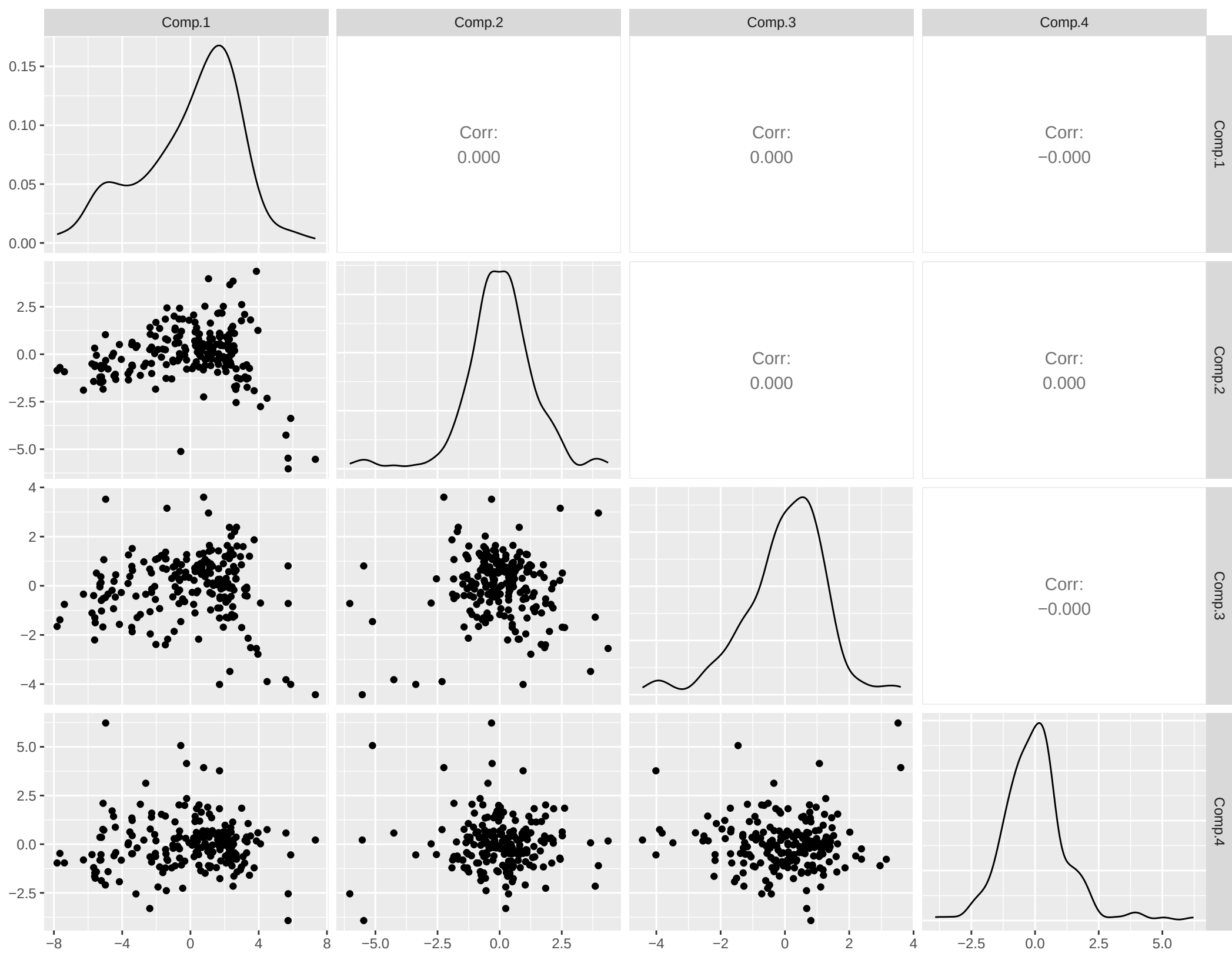


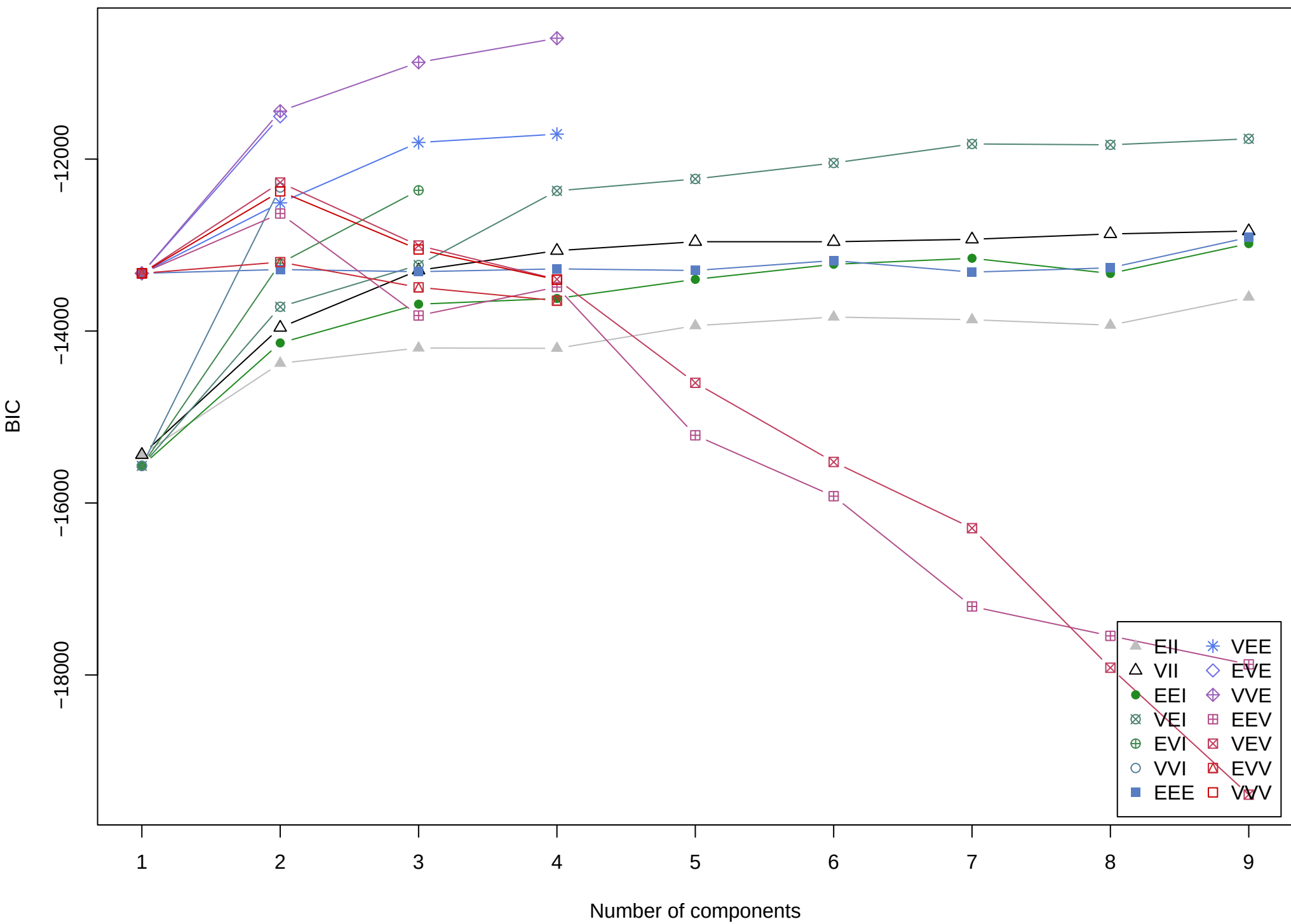
Income Group (real)



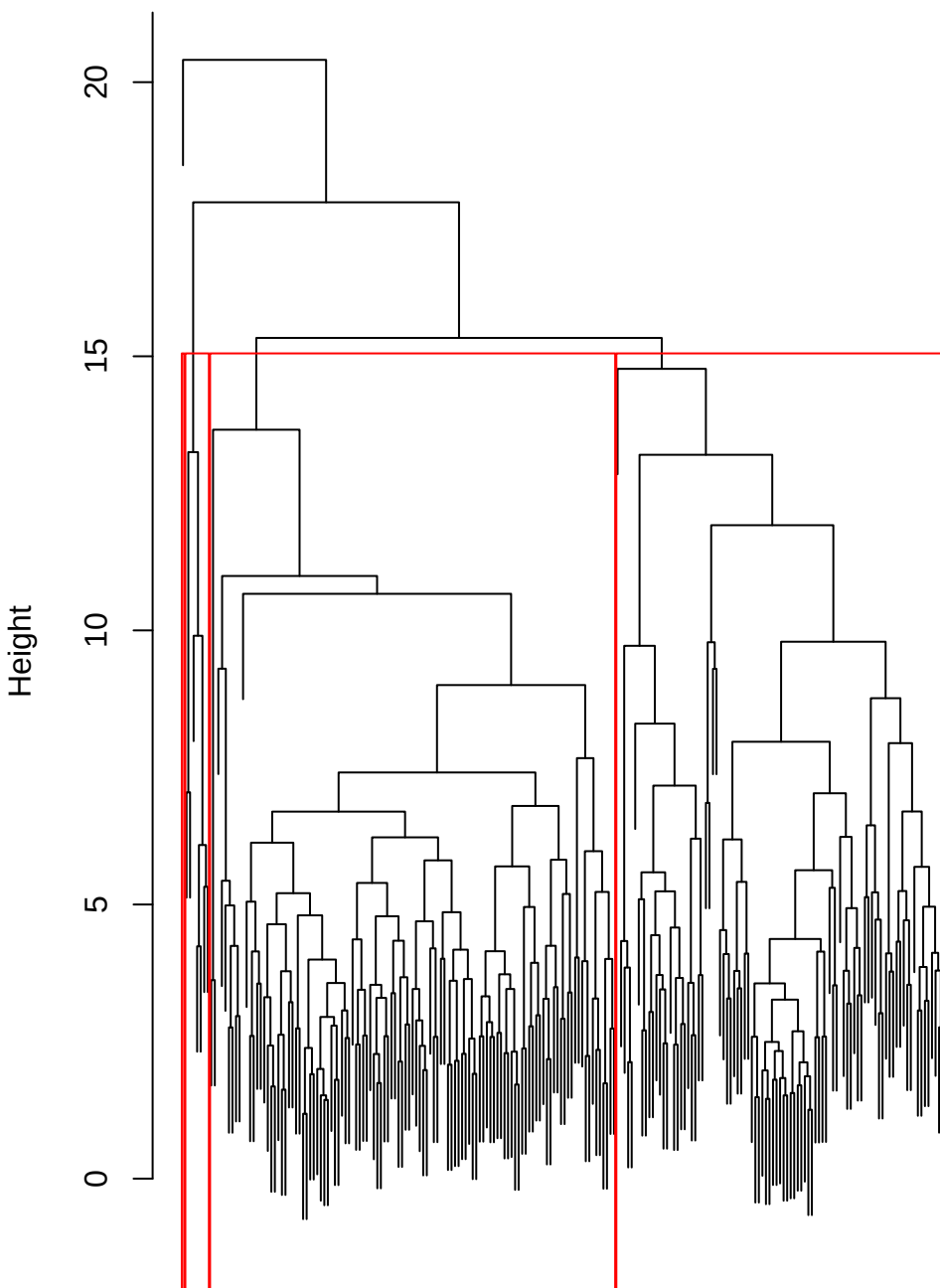
K-Means K=4



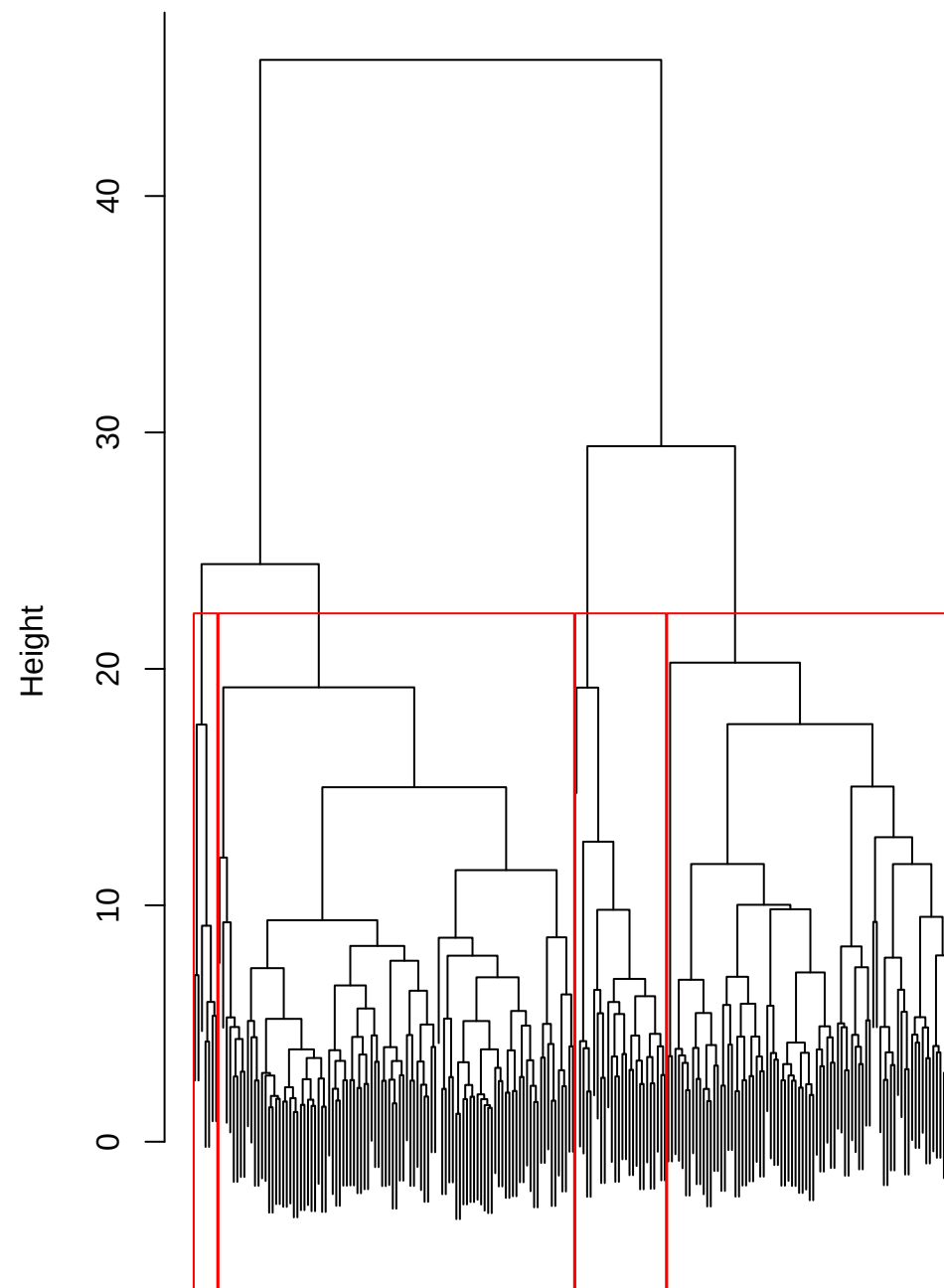




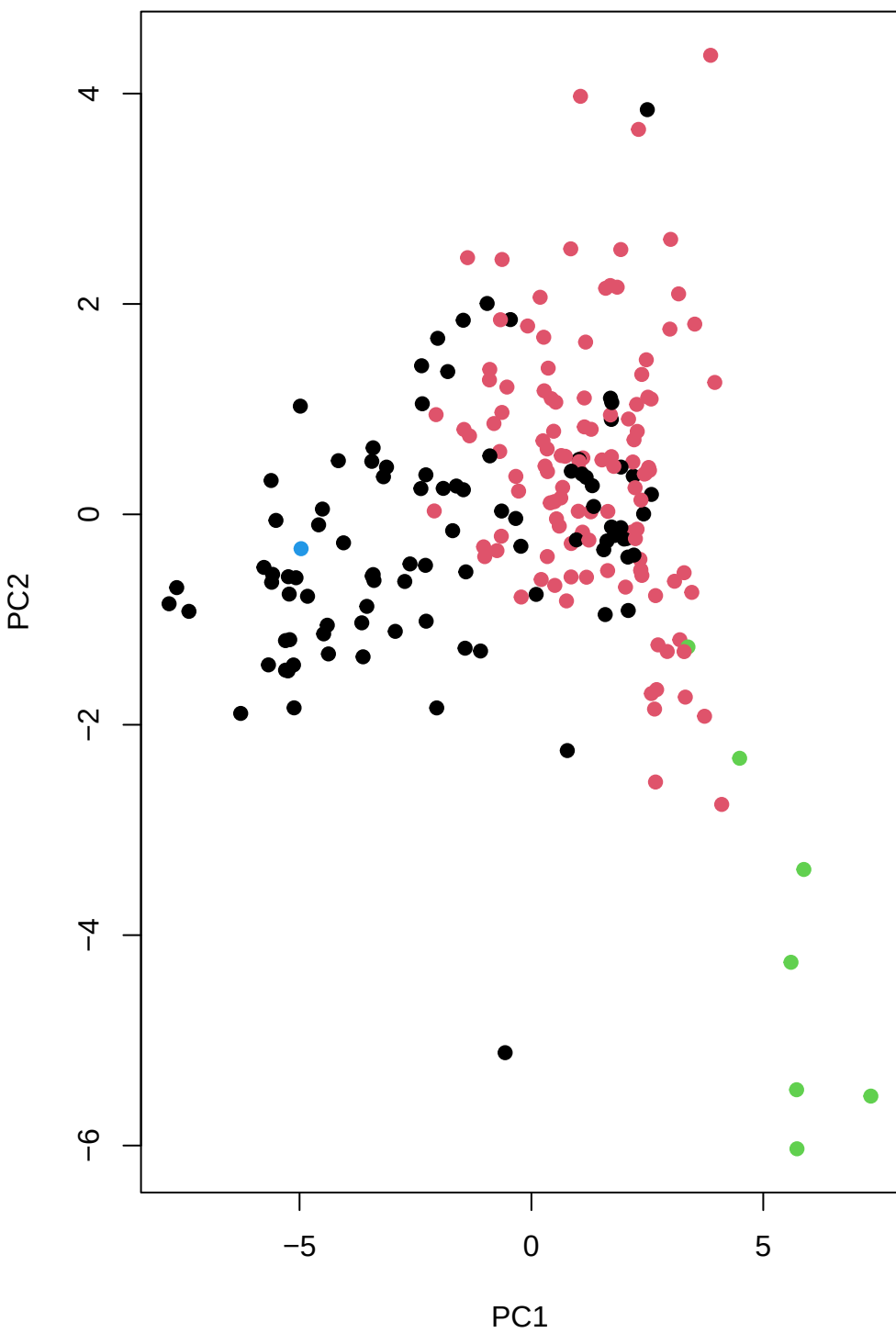
Dendrogram – Complete linkage



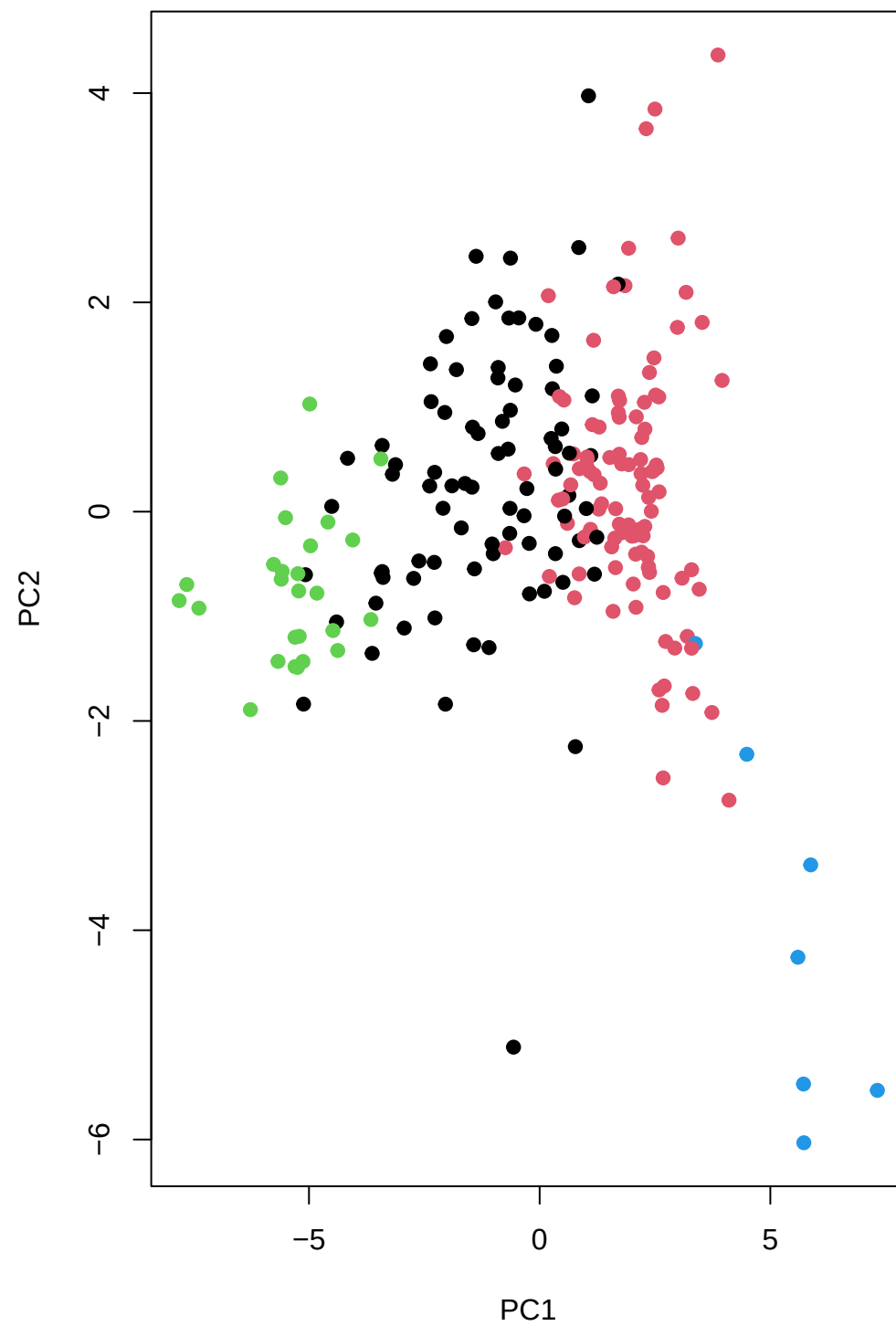
Dendrogram – Ward.D2



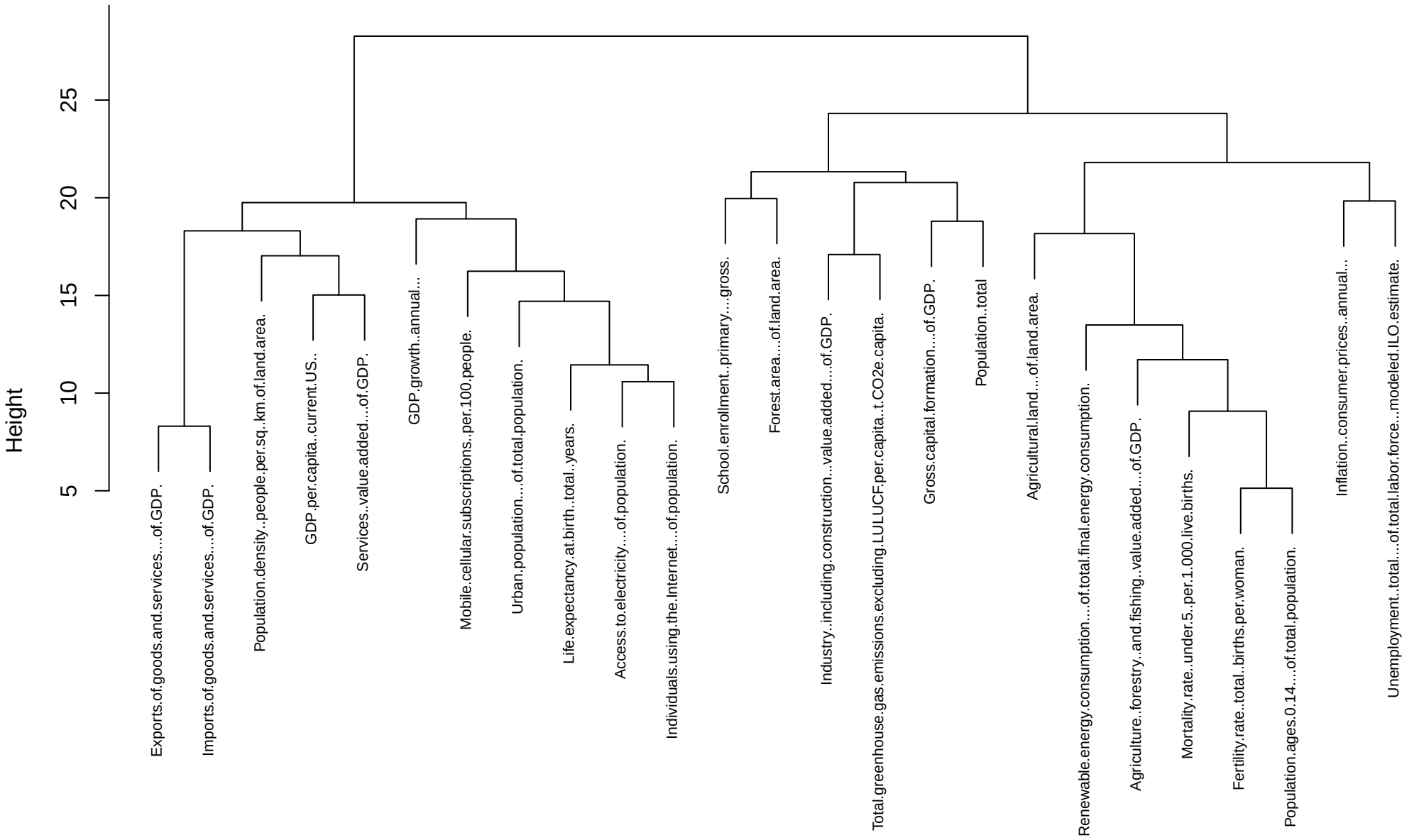
Hierarchical – Complete



Hierarchical – Ward.D2

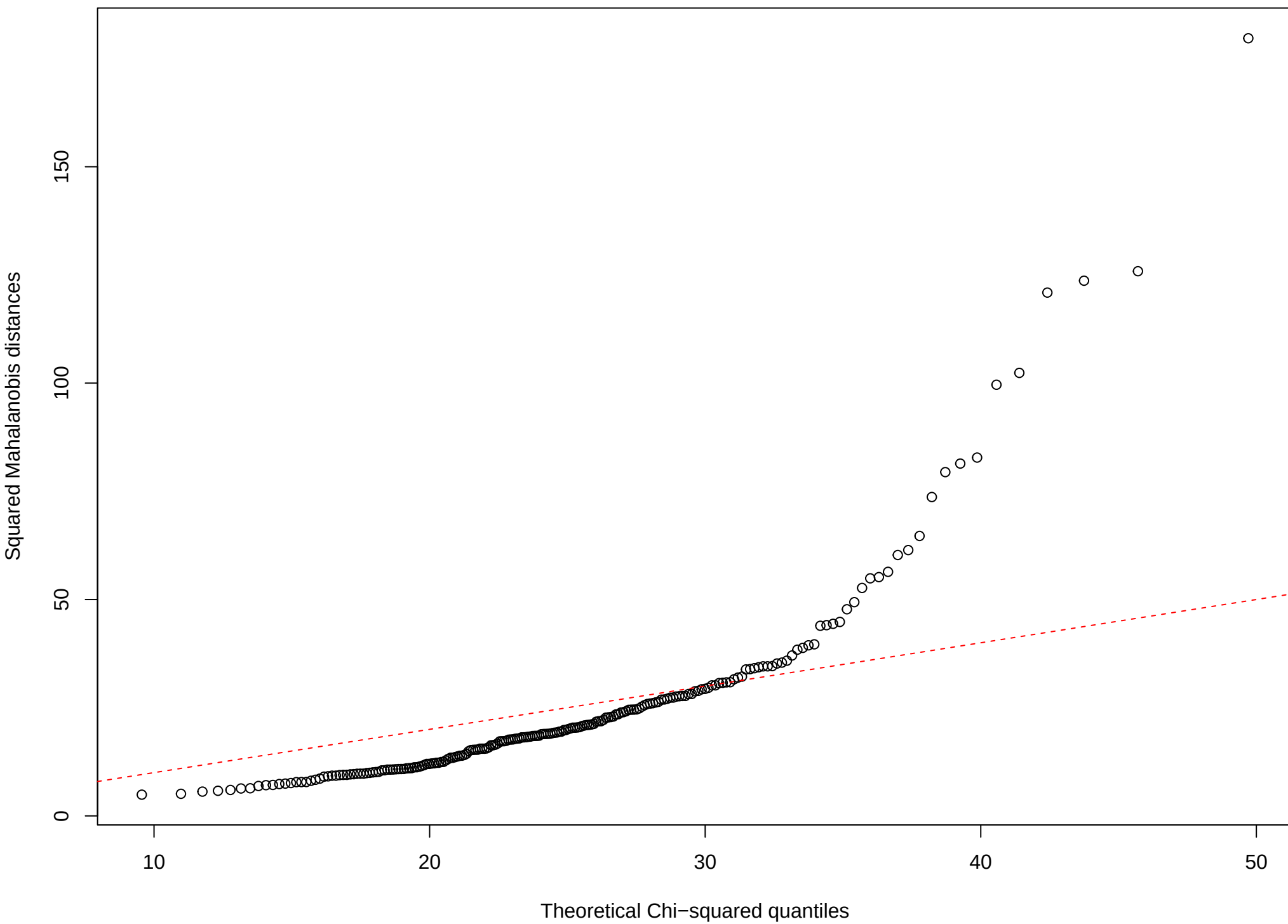


Dendrogram of Variables

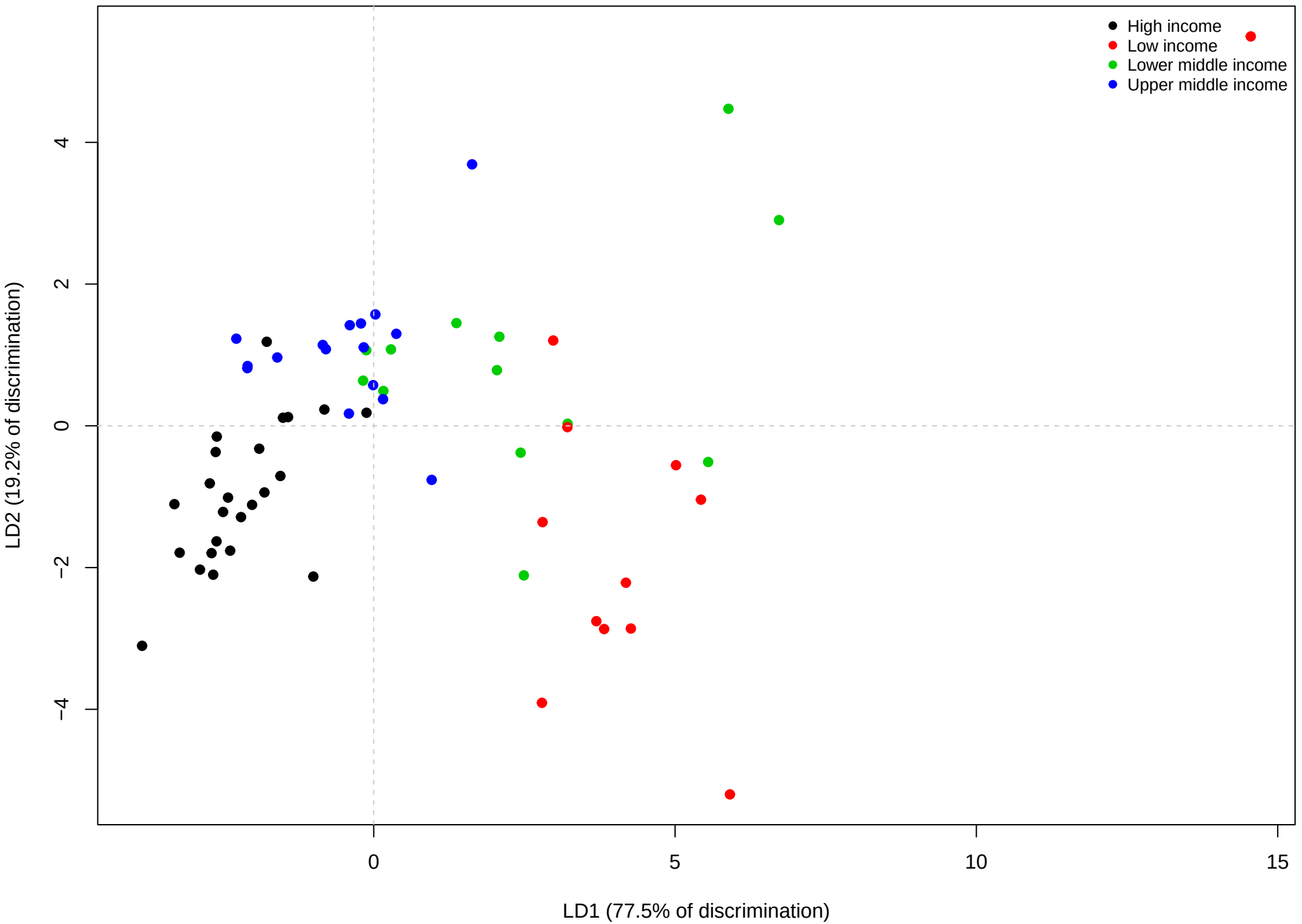


dist(t(dados_scaled))
hclust (*, "complete")

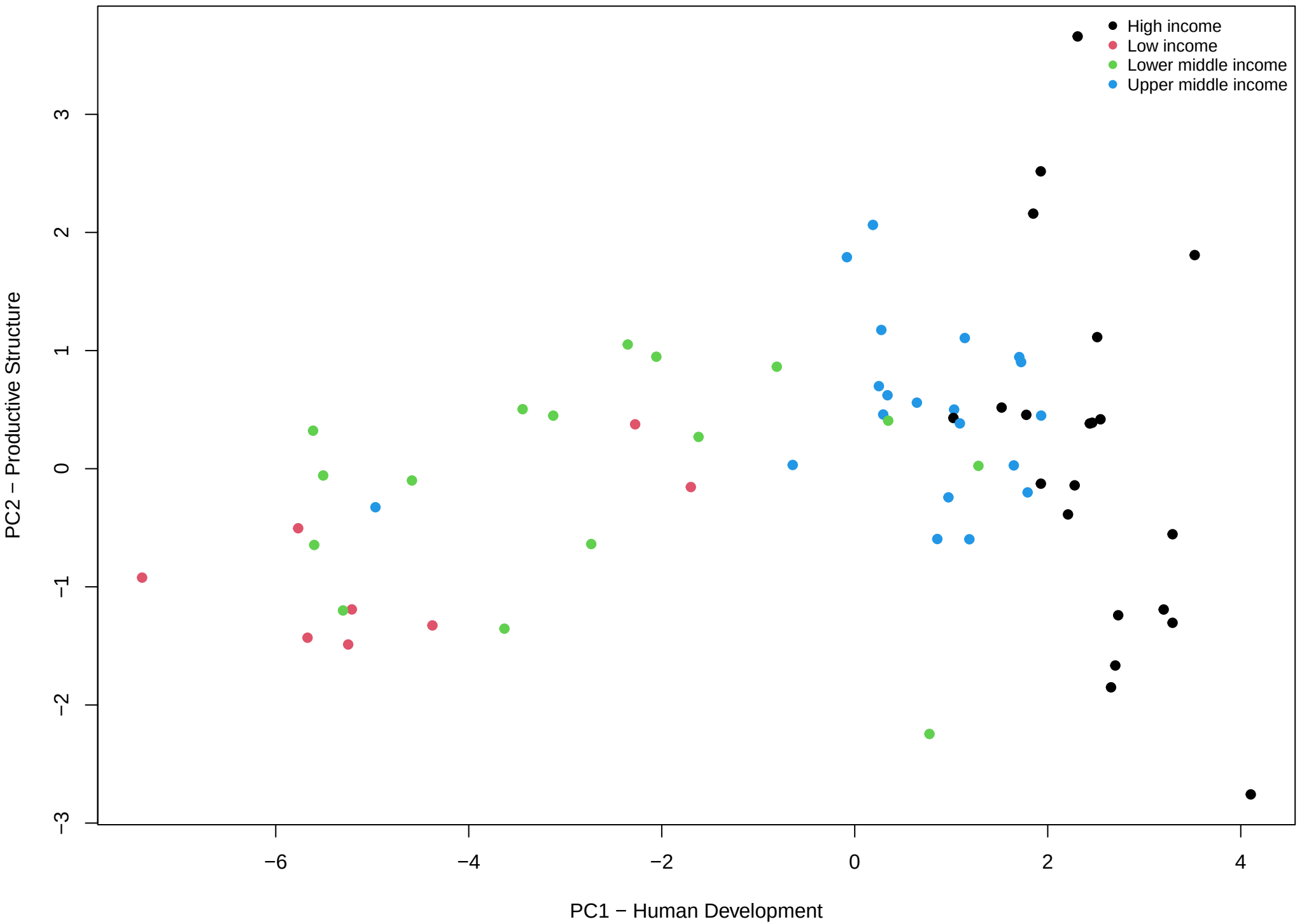
QQ-plot – Multivariate Normality Diagnostic



LDA – test set projection

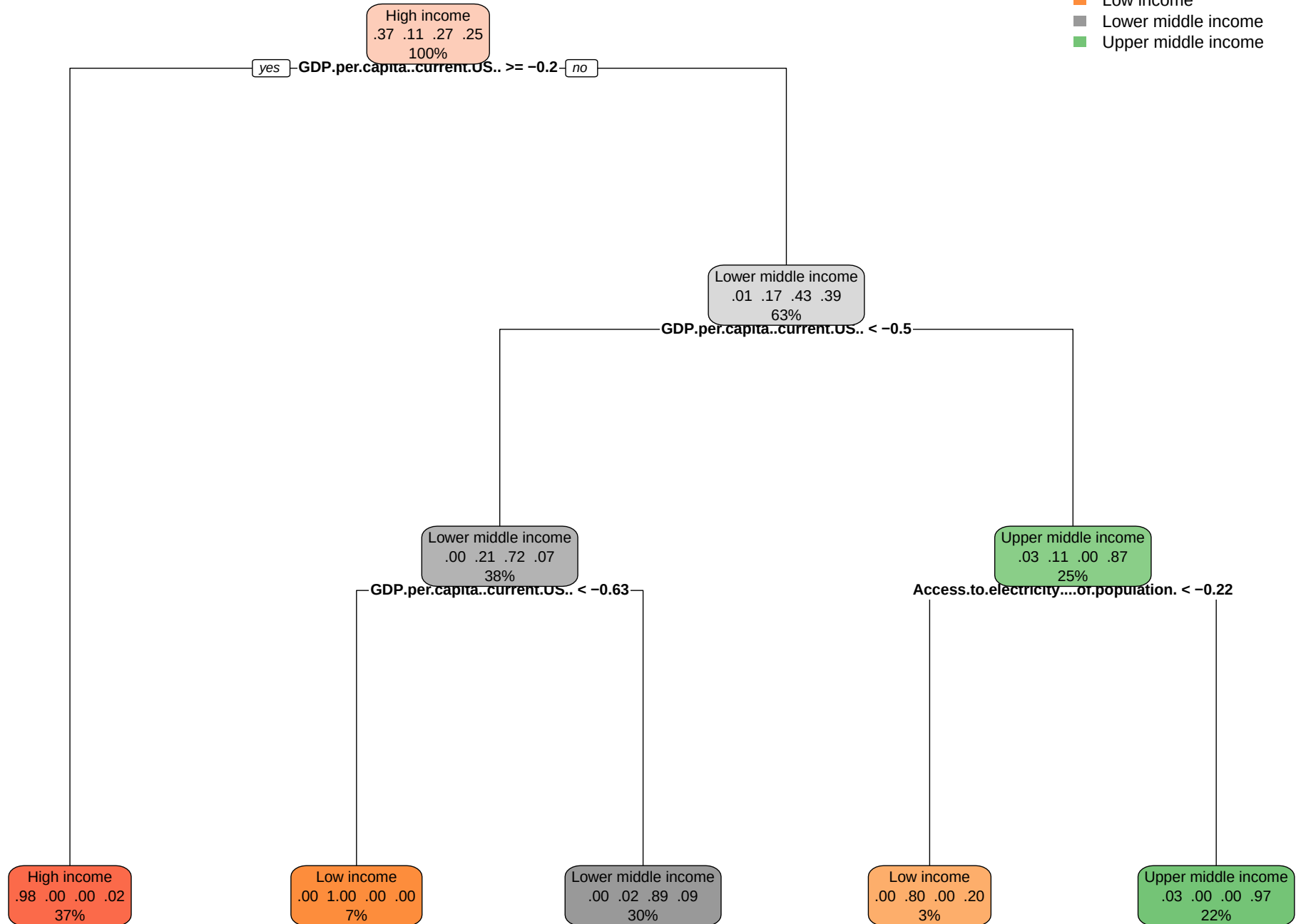


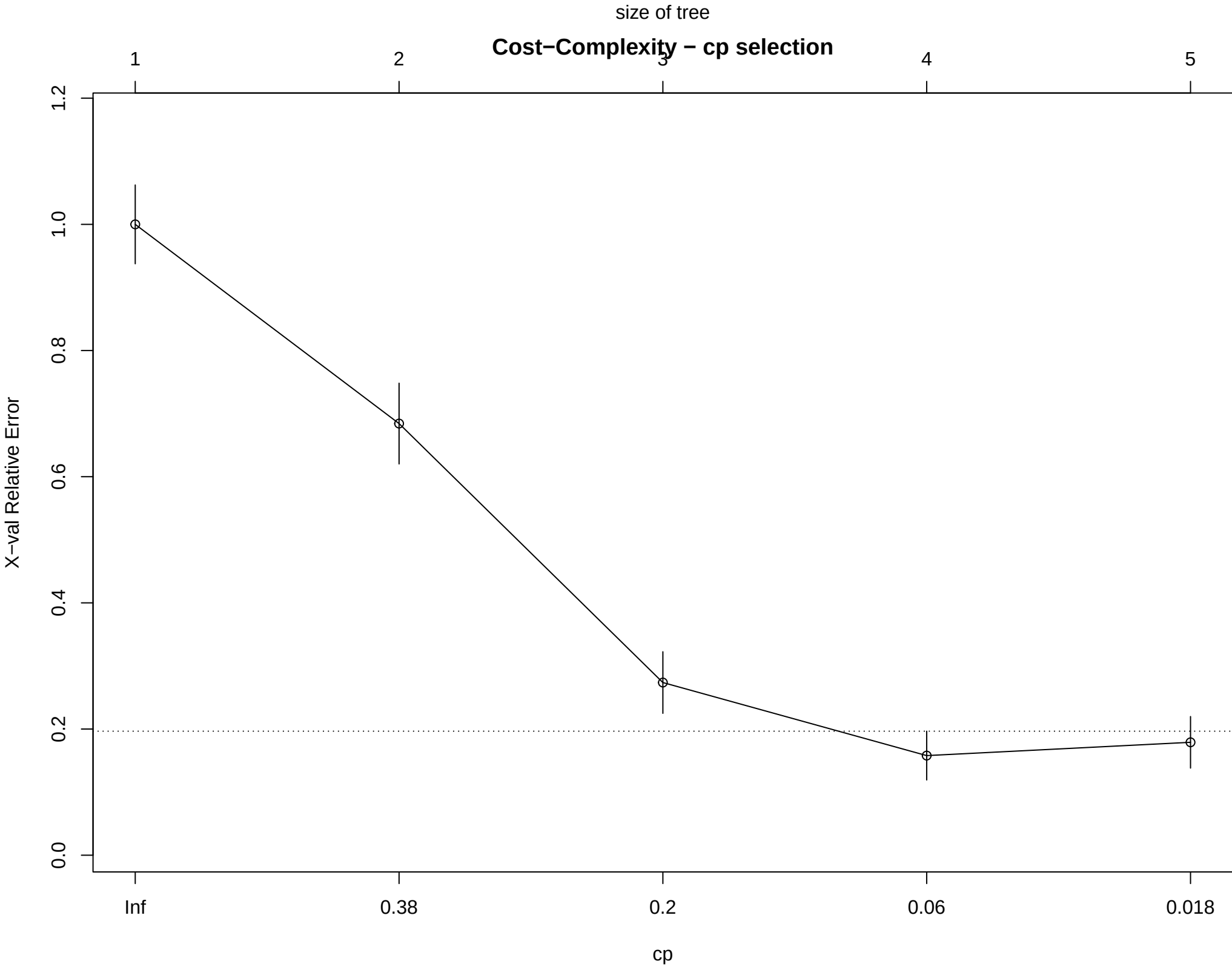
QDA – test set predictions (PC space)



CART Tree (before pruning)

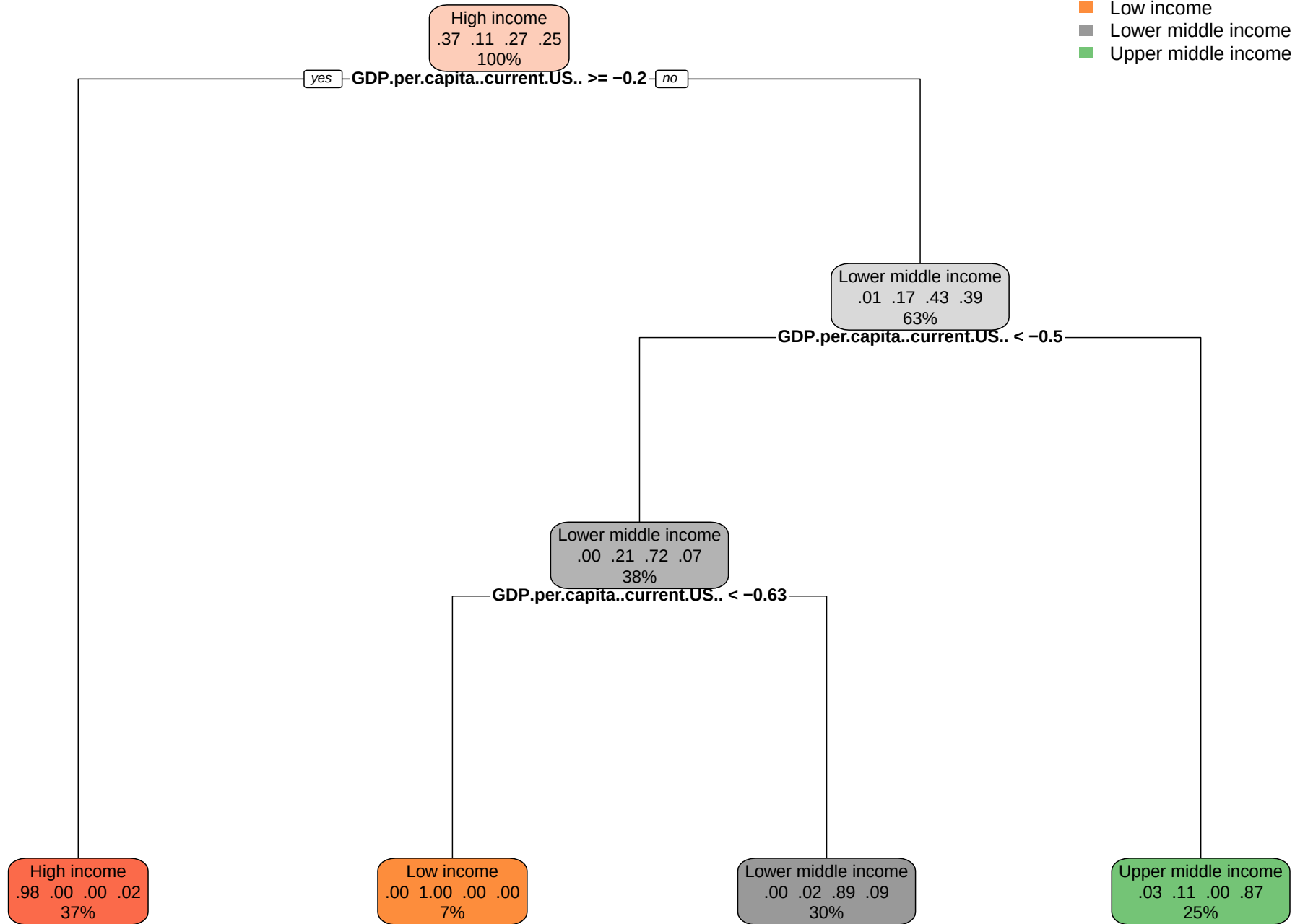
- High income
- Low income
- Lower middle income
- Upper middle income





CART Tree (after pruning)

- High income
- Low income
- Lower middle income
- Upper middle income



KNN – K selection by cross-validation

